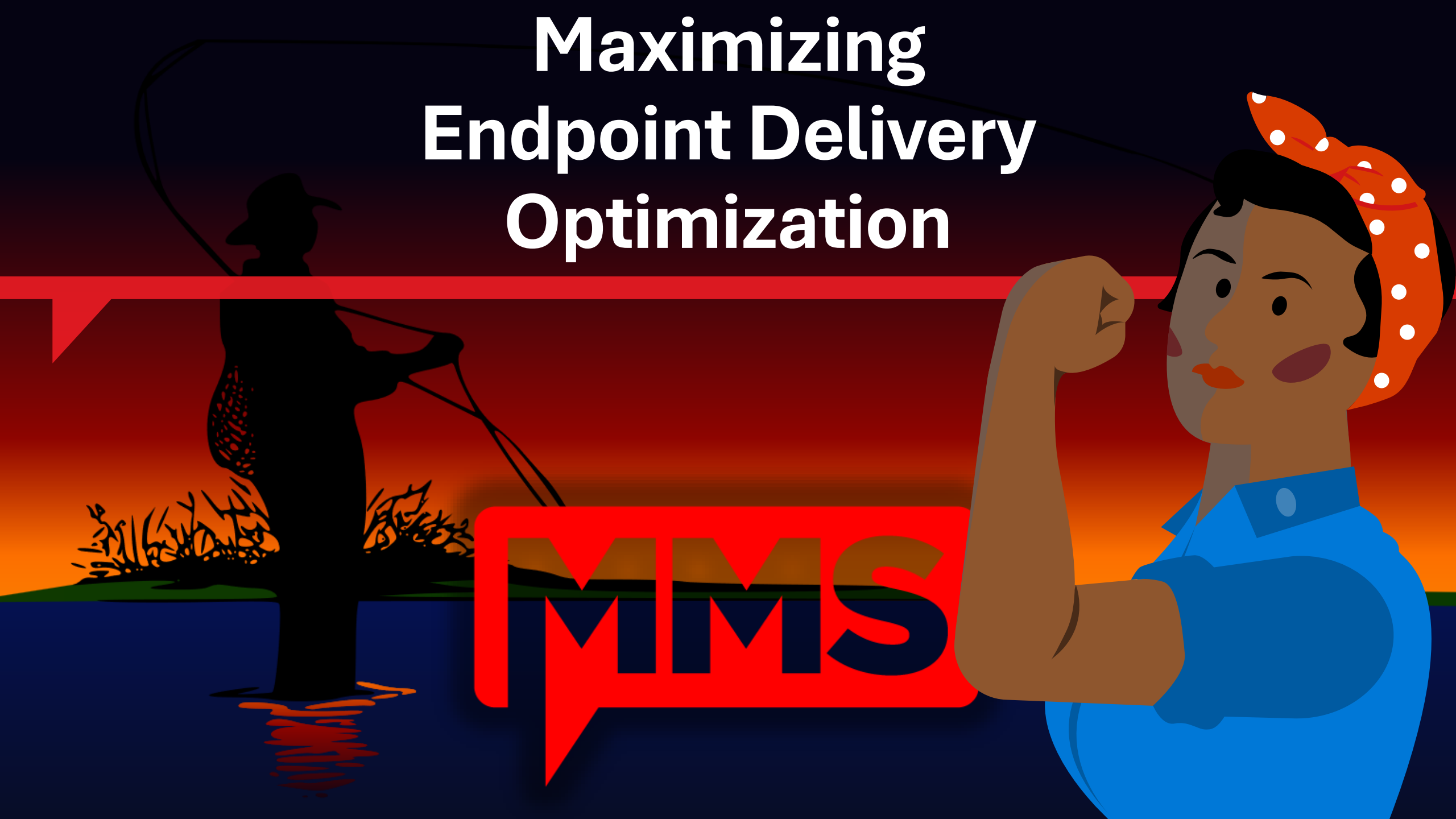


Maximizing Endpoint Delivery Optimization



Speakers



Andreas Hammarskjöld

Founder
2Pint Software

 @AndHammarskjold



Mattias Benninge

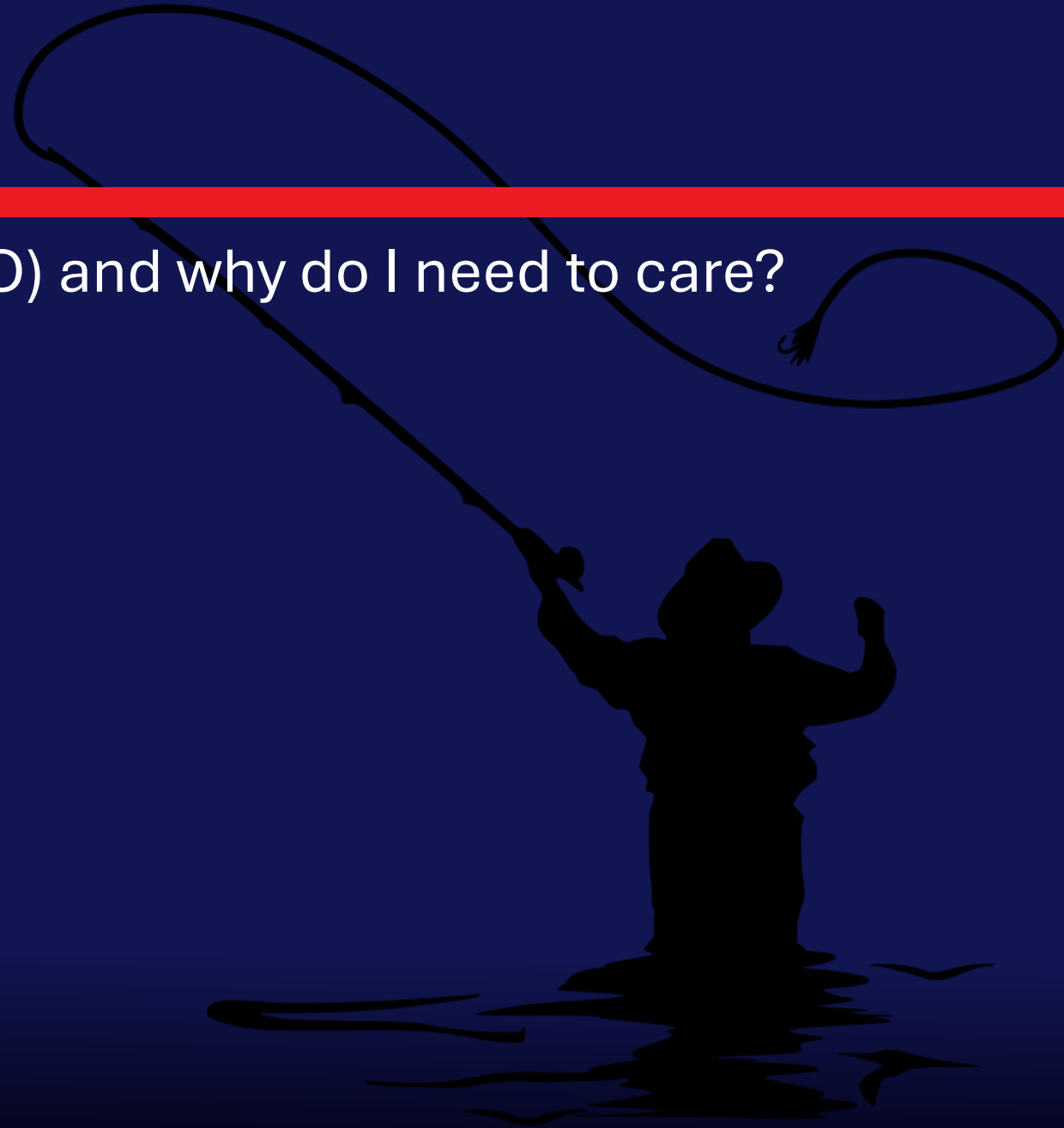
Principal Engineer
2Pint Software

 @matbg

Can't see our slides? Can't hear? Need to repeat the question? Call us out!

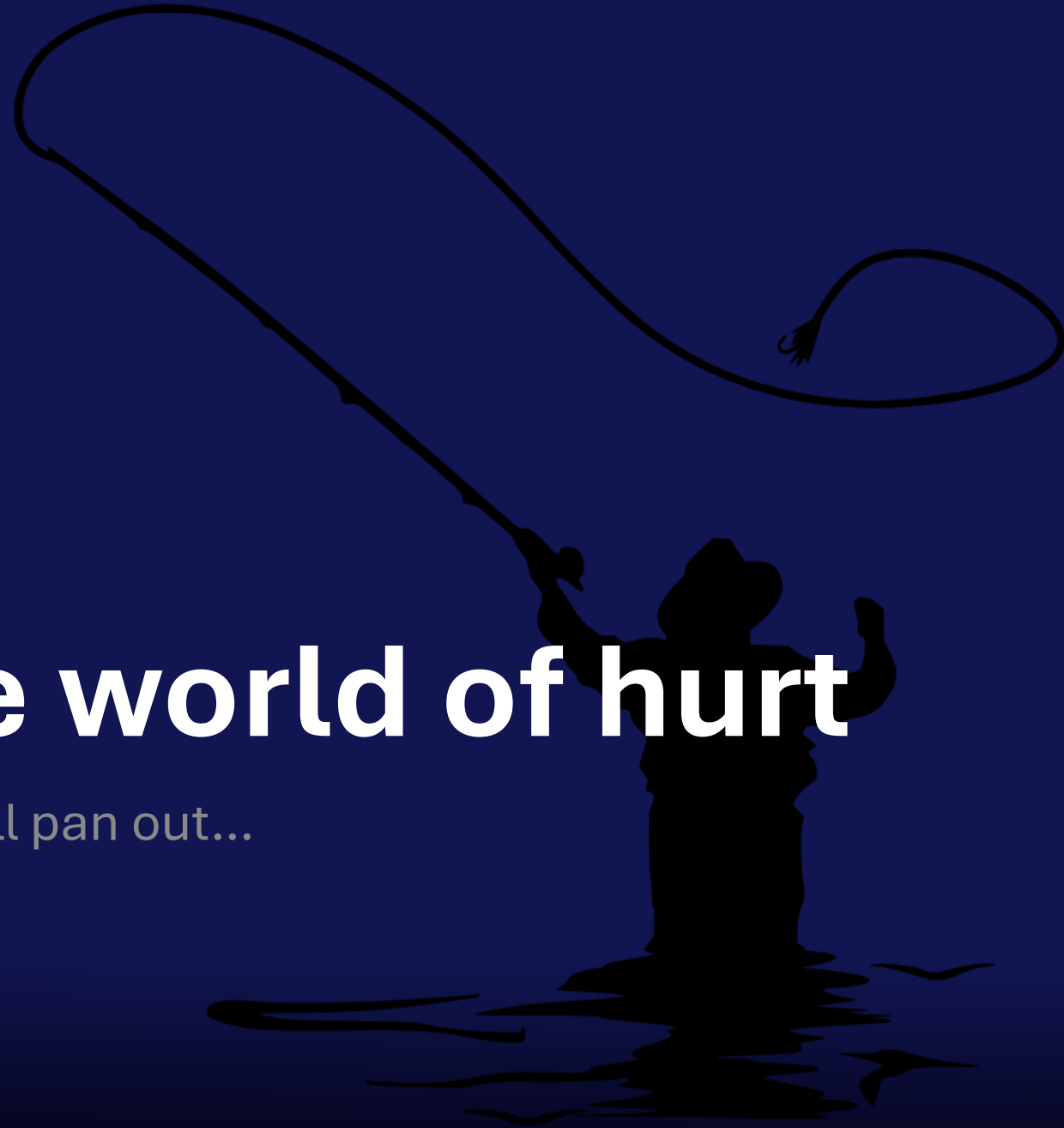
Agenda

- ◆ What is Delivery Optimization (DO) and why do I need to care?
- ◆ What are the key aspects of DO
 - ◆ Downloading
 - ◆ Peering
 - ◆ Caching
- ◆ What to keep track off in 2026
- ◆ How to maximize efficiency
- ◆ Keeping an eye on the stats
- ◆ Q & A



Welcome to the world of hurt

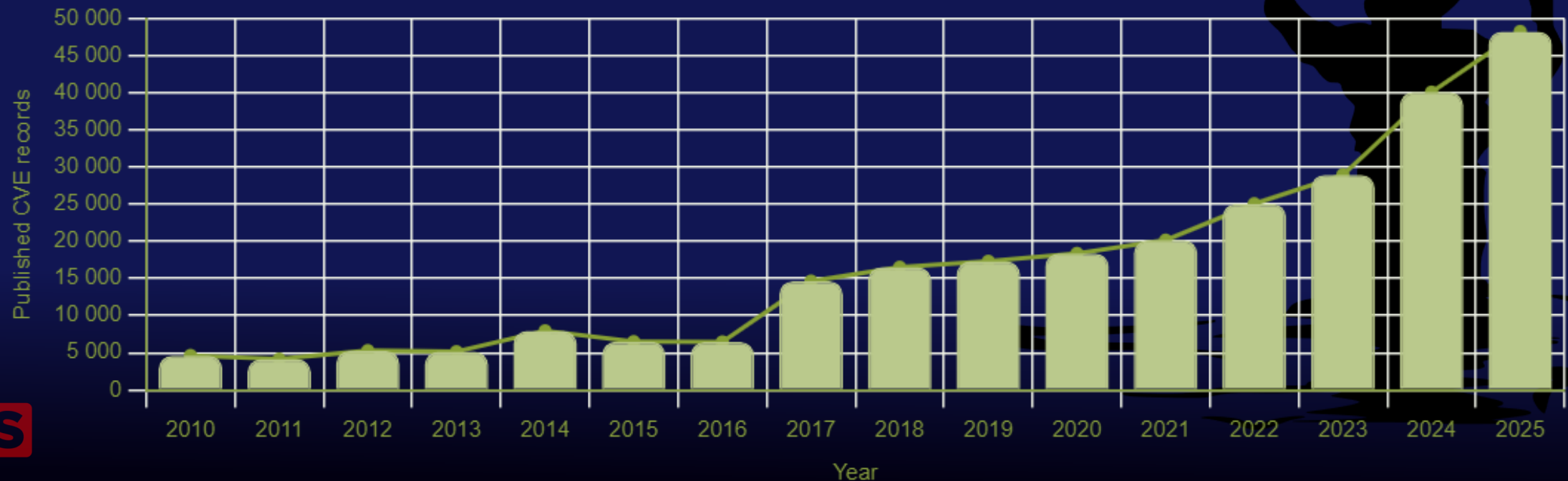
An insight into how the next few years will pan out...



Networks will be busier...

Year	Published / Received CVEs
2023	28,961
2024	40,077
2025	48,244
2026 (YTD)	21,044

Published CVE records per year (2010–2025)



Instant 0 days

No more patch
Tuesday?

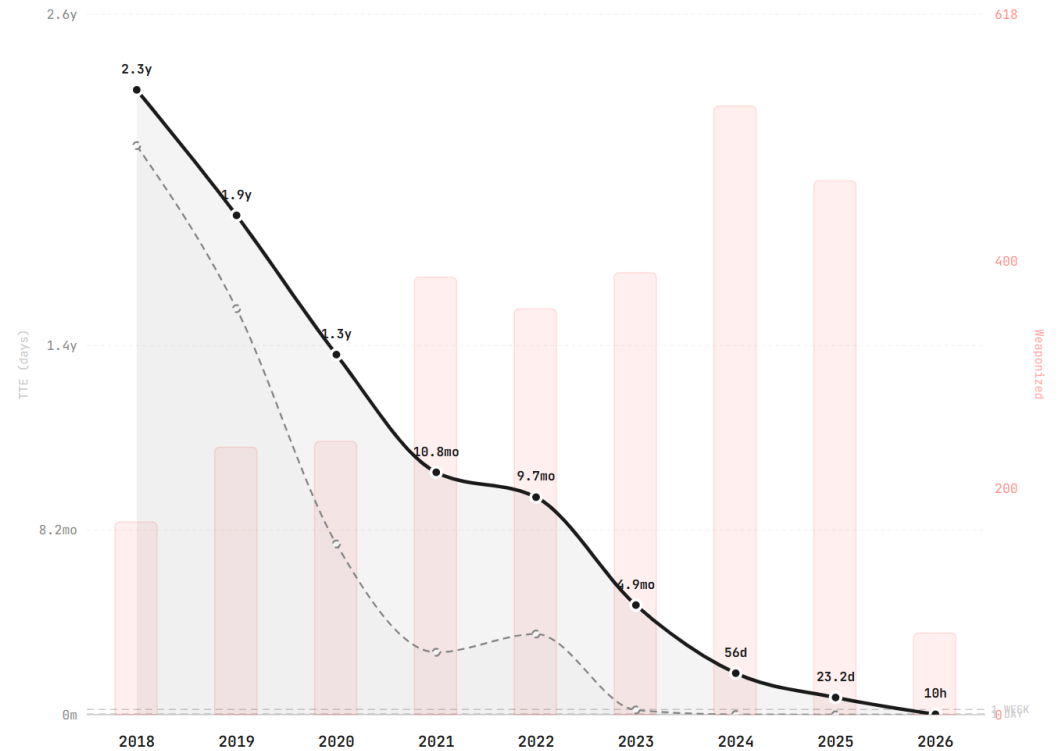
Speed is of essence!

ISV's will push/update
much faster!

From Vulnerability to Exploitation

TTE (Time-to-Exploit) measures the gap between CVE disclosure and confirmed exploitation

— Mean TTE (10% trimmed, days) - - - Median TTE (days) ■ Weaponized Exploits (count)

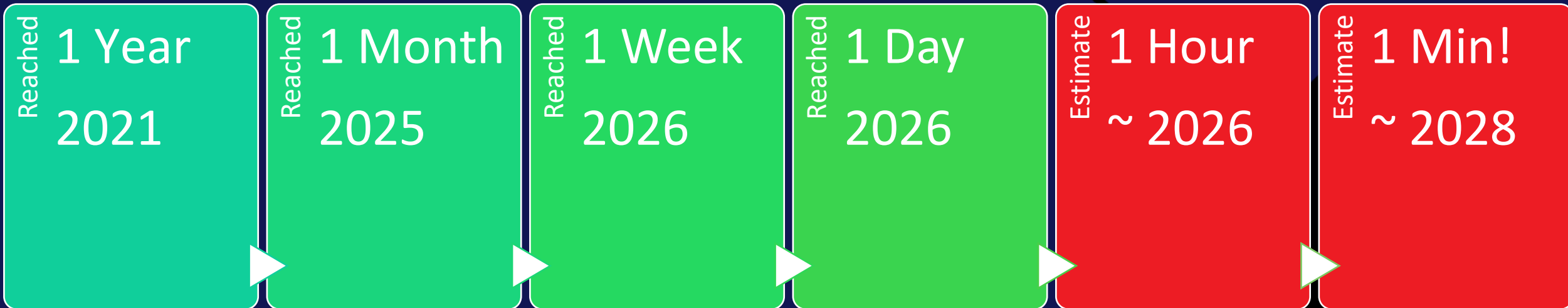


Based on 3 532 CVE-exploit pairs from trusted sources (CISA KEV, VulnCheck KEV & XDB)

• zerodayclock.com



Time-to-Exploit Milestones



<https://zerodayclock.com/>

More and more patches!

Size is going up!

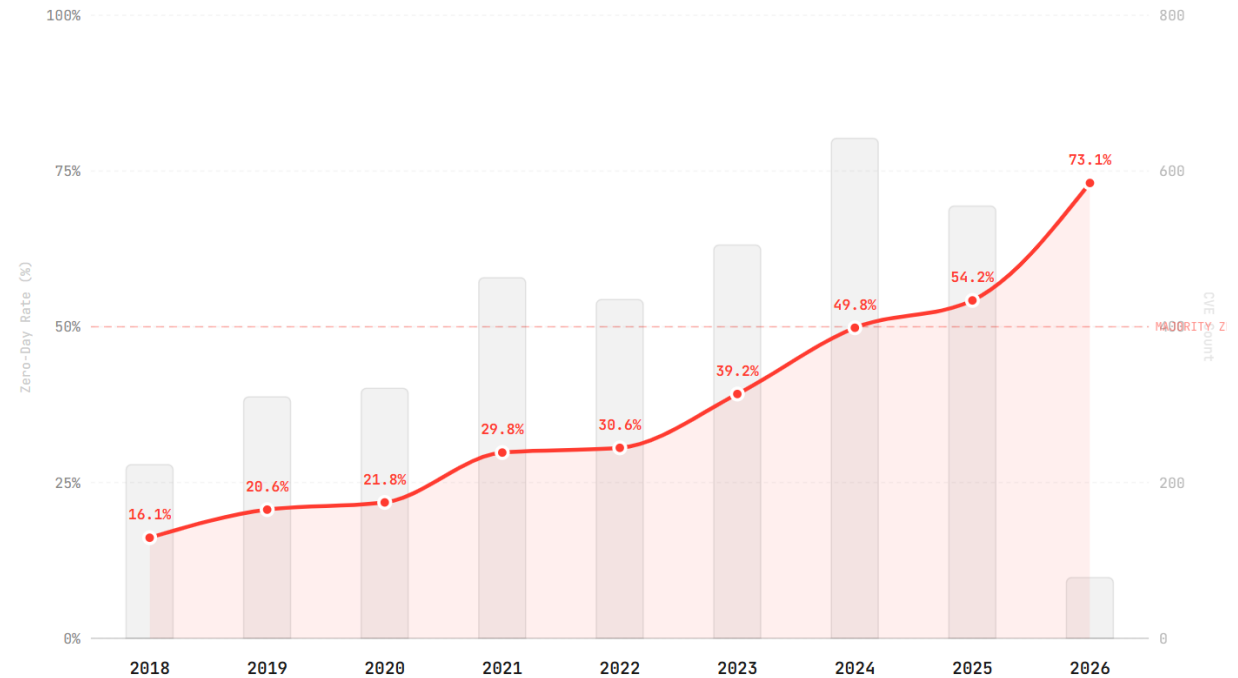
Issues is going up!

We need to patch faster and faster as the content is also getting bigger, bigger and bigger...

Zero-Day Rate

Percentage of exploited CVEs where exploitation occurred before or on the day of disclosure (TTE ≤ 0)

— Zero-Day Rate (% of exploited) ■ Exploited CVEs (count)



73.1% of exploited CVEs in 2026 are zero-days, up from 16.1% in 2018

zerodayclock.com

Know your audience!

Who is using Delivery Optimization today?



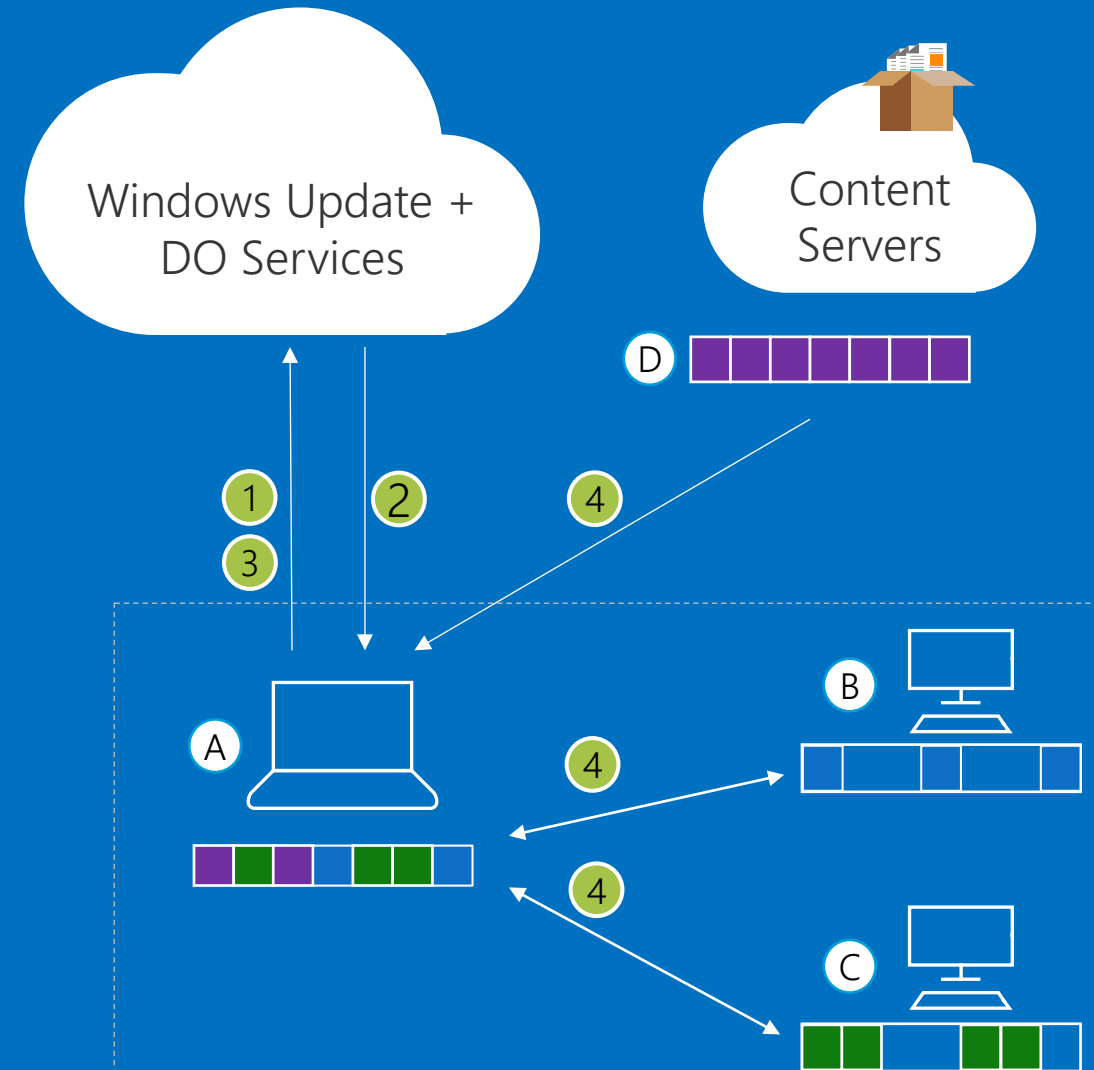
What is DO? How hard can it be?

- ◆ Download/caching/peering technology
 - ◆ It's a reliable HTTP download system for Windows that can fetch updates/apps from more than just the internet "HTTP source".
- ◆ Shares content locally (optional)
 - ◆ It can use peer-to-peer and/or a dedicated cache server (Microsoft Connected Cache) so multiple devices don't all pull the same large packages over the WAN.
- ◆ Falls back automatically
 - ◆ When configured for peers/cache, it tries those sources in parallel and seamlessly falls back to the HTTP source if the content can't be obtained from peers or cache

How does it work?

Step 2: Download

- (1) Client **A** checks for updates
- (2) Client **A** asks Windows Update service for download sources
- (3) Service provides Windows Update content server **D** and clients **B** and **C**
- (4) Client **A** requests specific small pieces from **D** and clients **B**, **C**
- (5) Source **B**, **C**, **D** fulfill the download request
- (6) Client **A** checks the hash of each piece, discard pieces where hashes don't match
- (7) Client **A** checks the hash of the entire file before installing



Key aspects to keep track off

Downloading

- Throttling – Speed
- Background vs Foreground

Peering

- Difference modes can give different efficiency

Caching

- Efficiency – once over the network

MCC Overview



DOINC

- Aka first MCC
- Sits on DPs



MCC for Windows

- Running on WSL

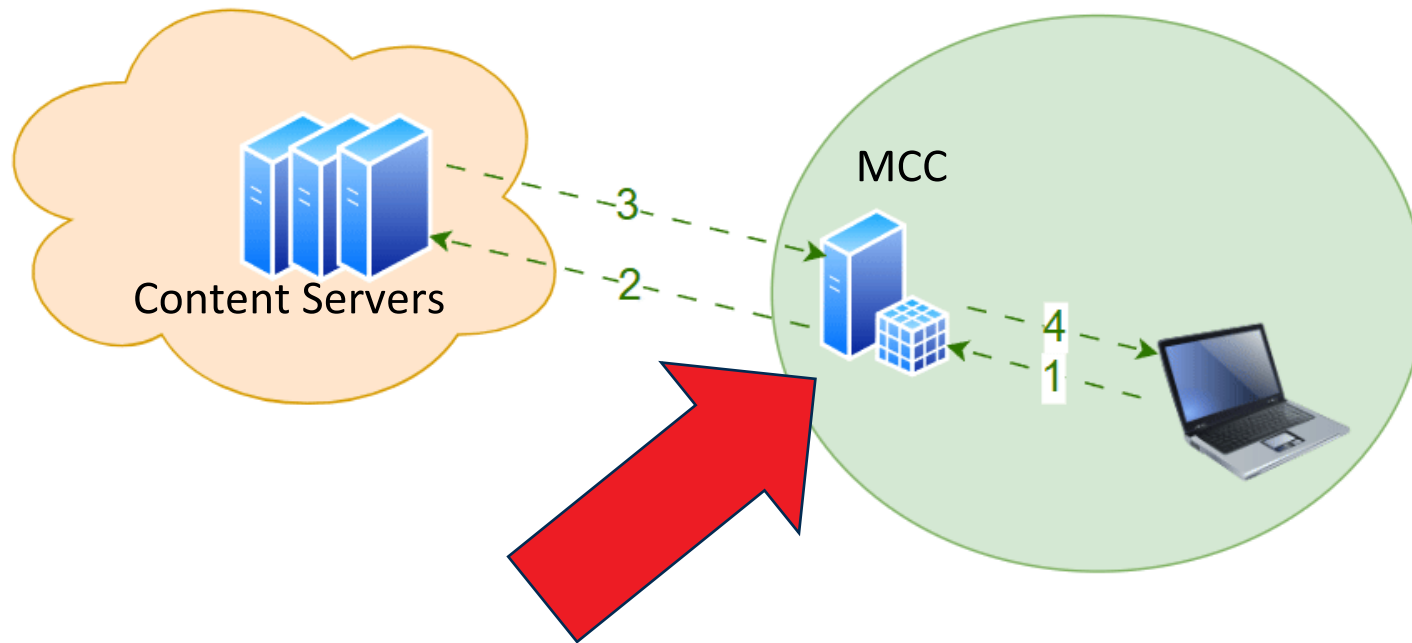


MCC for Linux

- Linux container

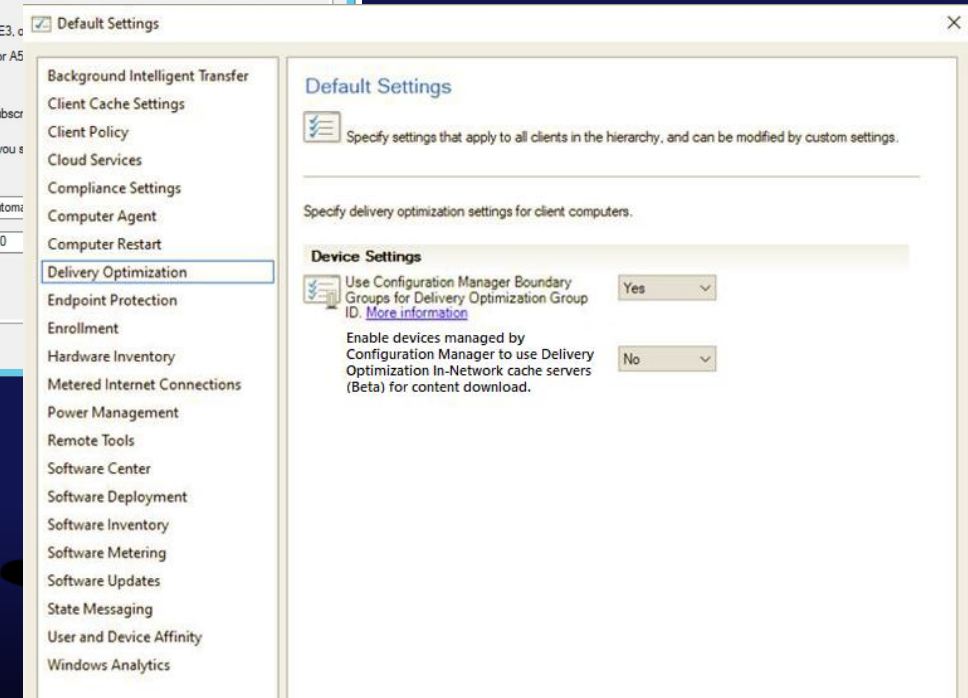
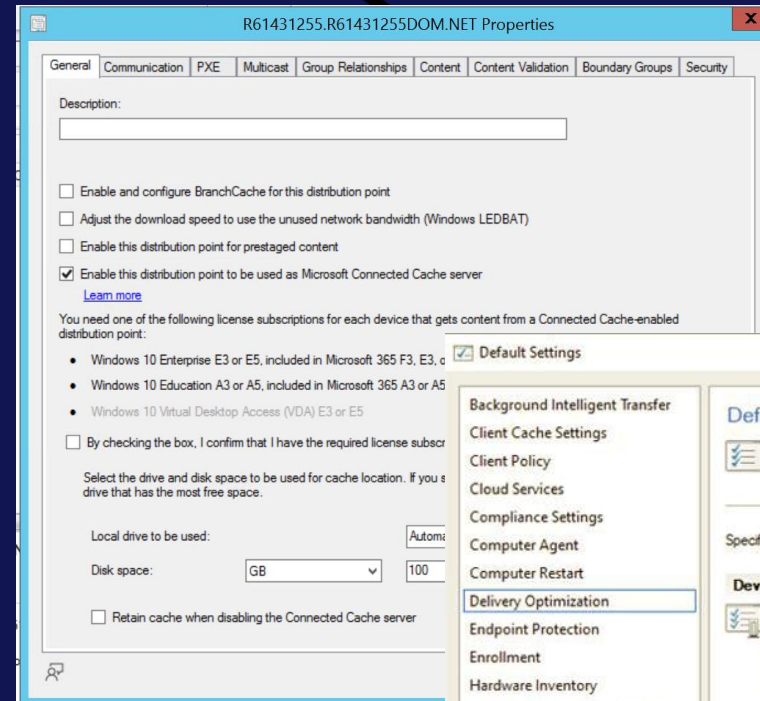


Simple MCC logic



Connected Cache for ConfigMgr

- ◆ Available in 2409, 2503, and 2509
- Enable with a few clicks
- Supports On-Premises, cloud-managed, and co-managed scenarios
- Supports https - [KB33247081](https://support.microsoft.com/kb/33247081)



MCC, “modern” DPs?

MCC (Microsoft Connect Cache) for Enterprise and Education

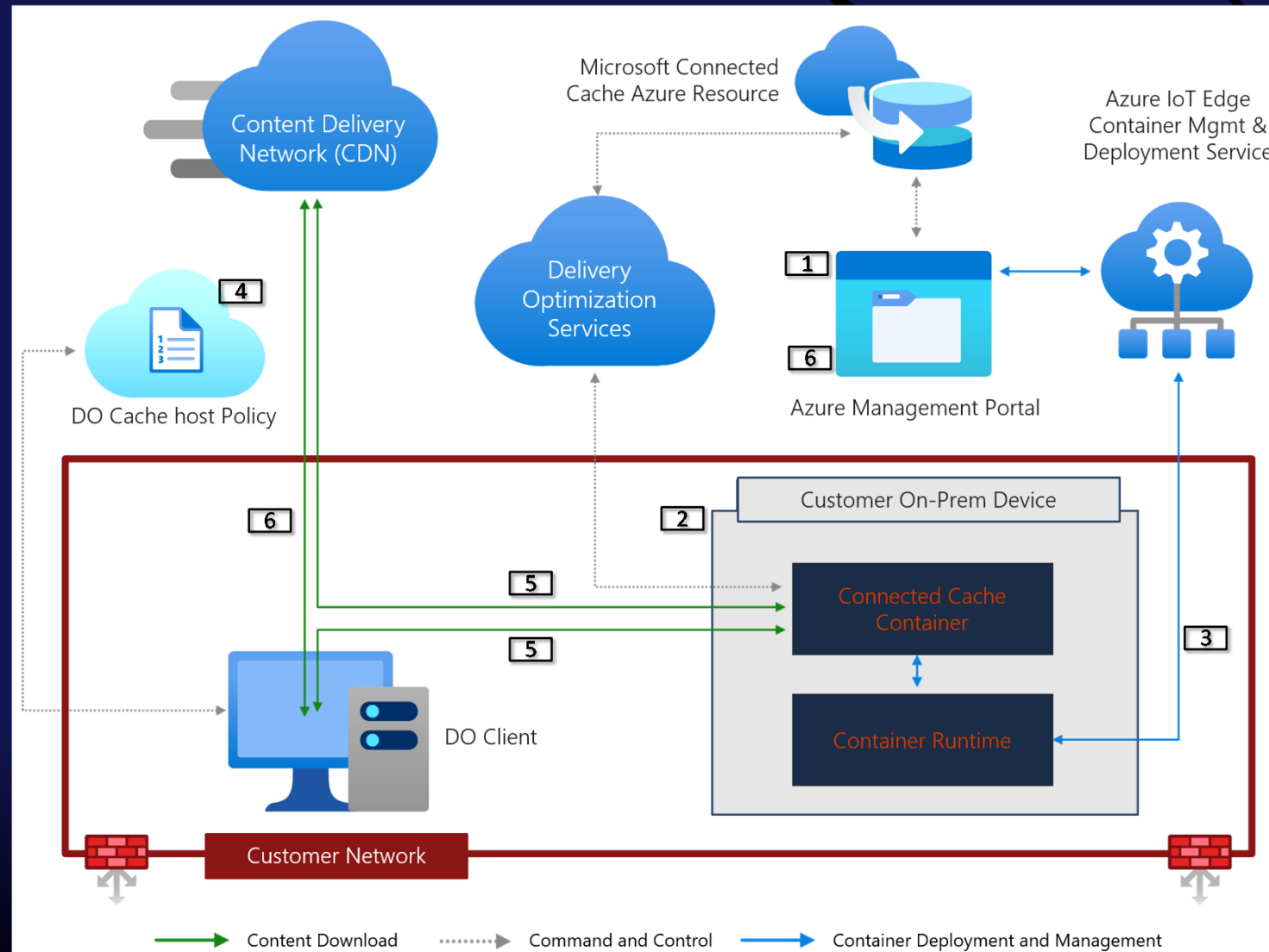
- ◆ Now GA
- ◆ Runs in a Linux container – Native Ubuntu or RHEL or Windows 11 and Windows Server
- ◆ New version that enabled https support. IMPORTANT!

*“Microsoft Intune will soon enforce HTTPS-only delivery for all managed Win32 applications. **Cache nodes without HTTPS support will be bypassed, and clients will fall back to CDN delivery.**”*

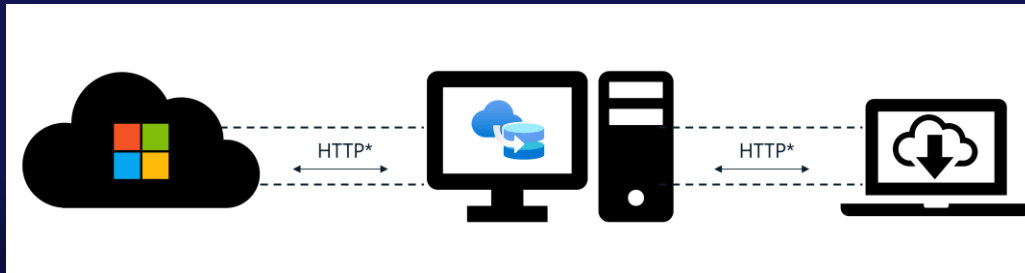
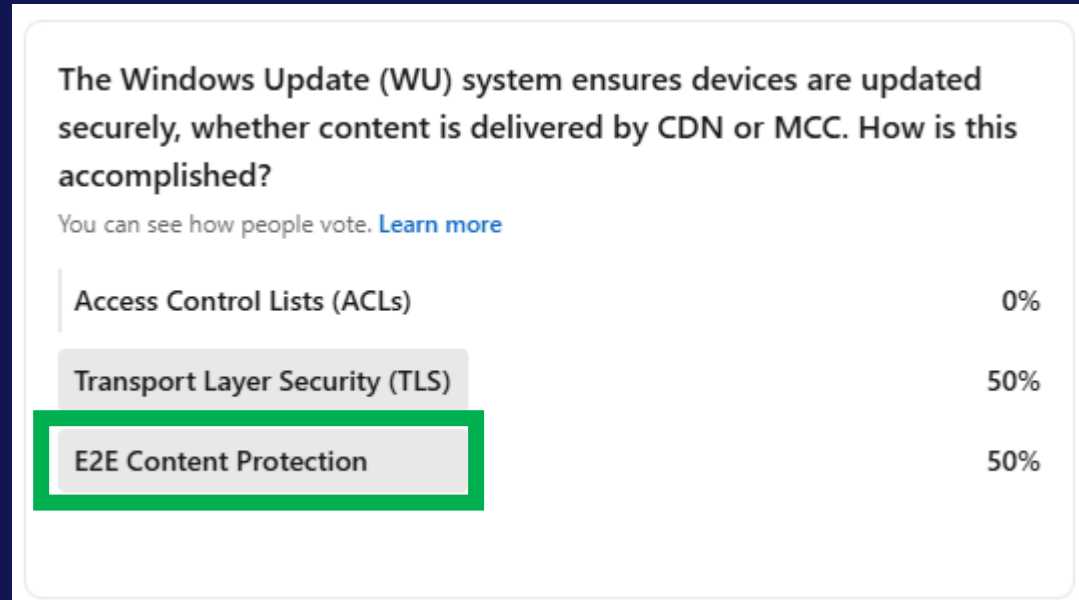
<https://learn.microsoft.com/en-us/windows/deployment/do/mcc-ent-https-overview>

NOTE: Use P2P where you can, otherwise you might end up with a bunch of “modern” DPs (MCC) in each locations that you tried so hard to get rid of when you moved off ConfigMgr..

How does MCC for Enterprise and Education (aka standalone) work?



Connected Cache & Windows Update content security



- Regardless of download source, the DO client on each Windows device verifies the authenticity and integrity of downloaded content using its metadata hash and content hash
- The downloaded content's signature is verified before it is installed.
- <https://learn.microsoft.com/en-us/windows/deployment/do/mcc-ent-secure-content-delivery>

Configuring MCC for Enterprise and Education to deliver https

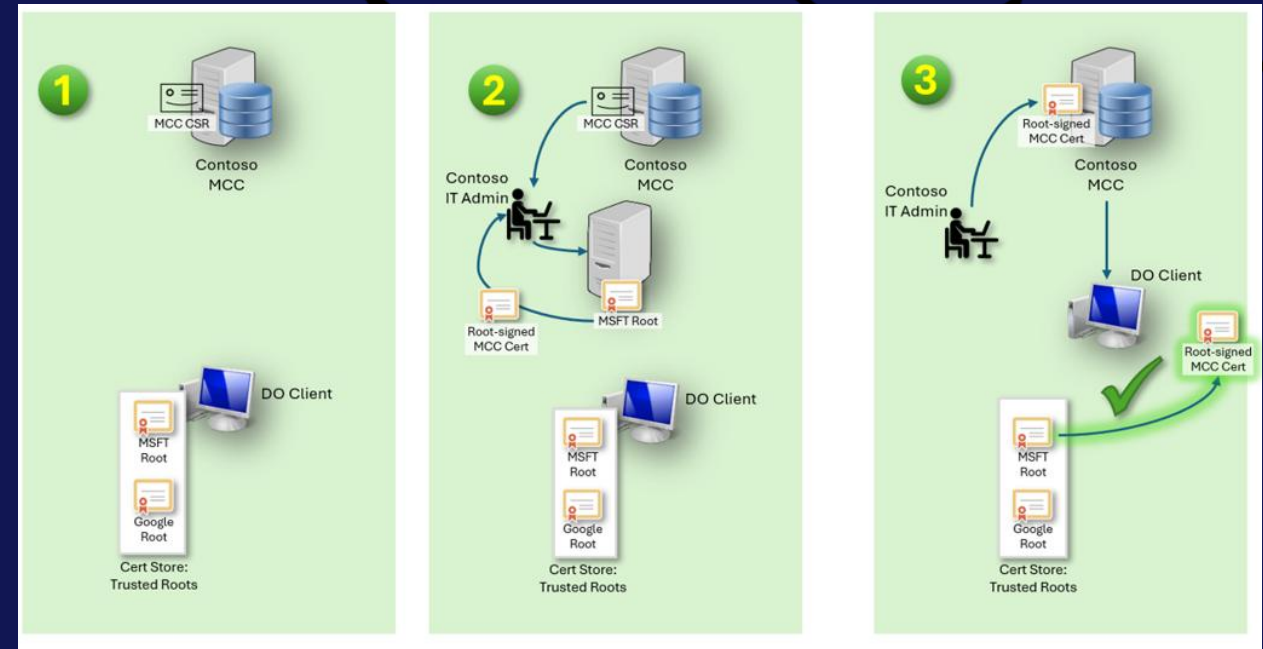
Generate a Certificate Signing Request on Connected Cache server

Sign CSR with your Certificate Authority

Import signed certificate to Connected Cache server

Ensure clients have the same trusted root certificate

Https content cached and delivered to clients



Verify functionality

- ◆ There are three possible results to determine whether HTTPS downloads from Microsoft Connected Cache are functioning properly:
 - ◆ “CDN” and “DOINC” bytes are equal value. This means 100% of the bytes came from cache.
 - ◆ “DOINC” bytes are zero. This means that 100% of the bytes came from CDN.
 - ◆ “CDN” greater than “DOINC” bytes. This means partial bytes came from cache server.

What about my proxy and MCC?

- ◆ MCC works with TLS proxies (Zscaler, Blue Coat, Netskope) without changing client proxy settings.

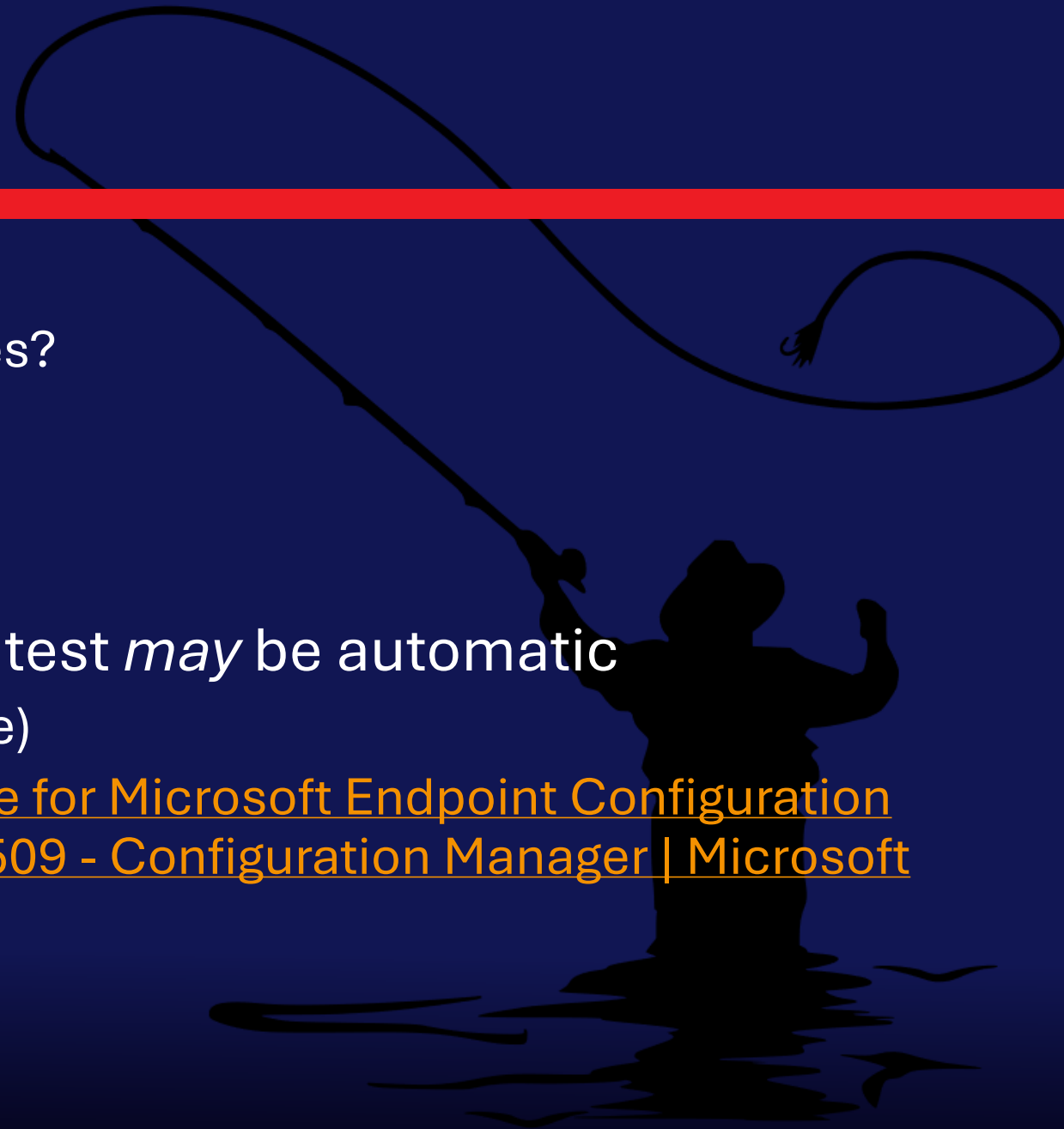
However... Allow *.prod.do.dsp.mp.microsoft.com to bypass your TLS-inspecting proxy entirely — intercepting this traffic breaks the certificate chain and will cause MCC deployment and operation to fail (true for peering as well).

- ◆ Forward proxies that cache content or expect absolute-form URLs (e.g., Squid defaults) — configure them to pass MCC traffic through to origin or exempt the MCC node from the proxy entirely.

- ◆ Upstream traffic still respects proxy policy - Cache nodes retrieve content from CDNs through the customer's configured proxy, maintaining inspection, logging, and egress controls ! Still need to allow CDN URLs !
- ◆ Local delivery bypasses proxy – Once cached, content is served locally over the LAN, reducing proxy load for repeated WAN downloads.
- ◆ Pay attention to the installation references to proxy

MCC Challenges

- ◆ Auto-Discovery of MCC
 - ◆ For Intune how do we do boundaries?
 - How can we set the group id?
 - How can we link to the right MCC?
- ◆ Proxies (earlier slide)
- ◆ MCC on ConfigMgr – Update to latest *may* be automatic
 - ◆ KB33247081 (No automatic update)
 - ◆ [Microsoft Connected Cache update for Microsoft Endpoint Configuration Manager version 2409, 2503 and 2509 - Configuration Manager | Microsoft Learn](#)



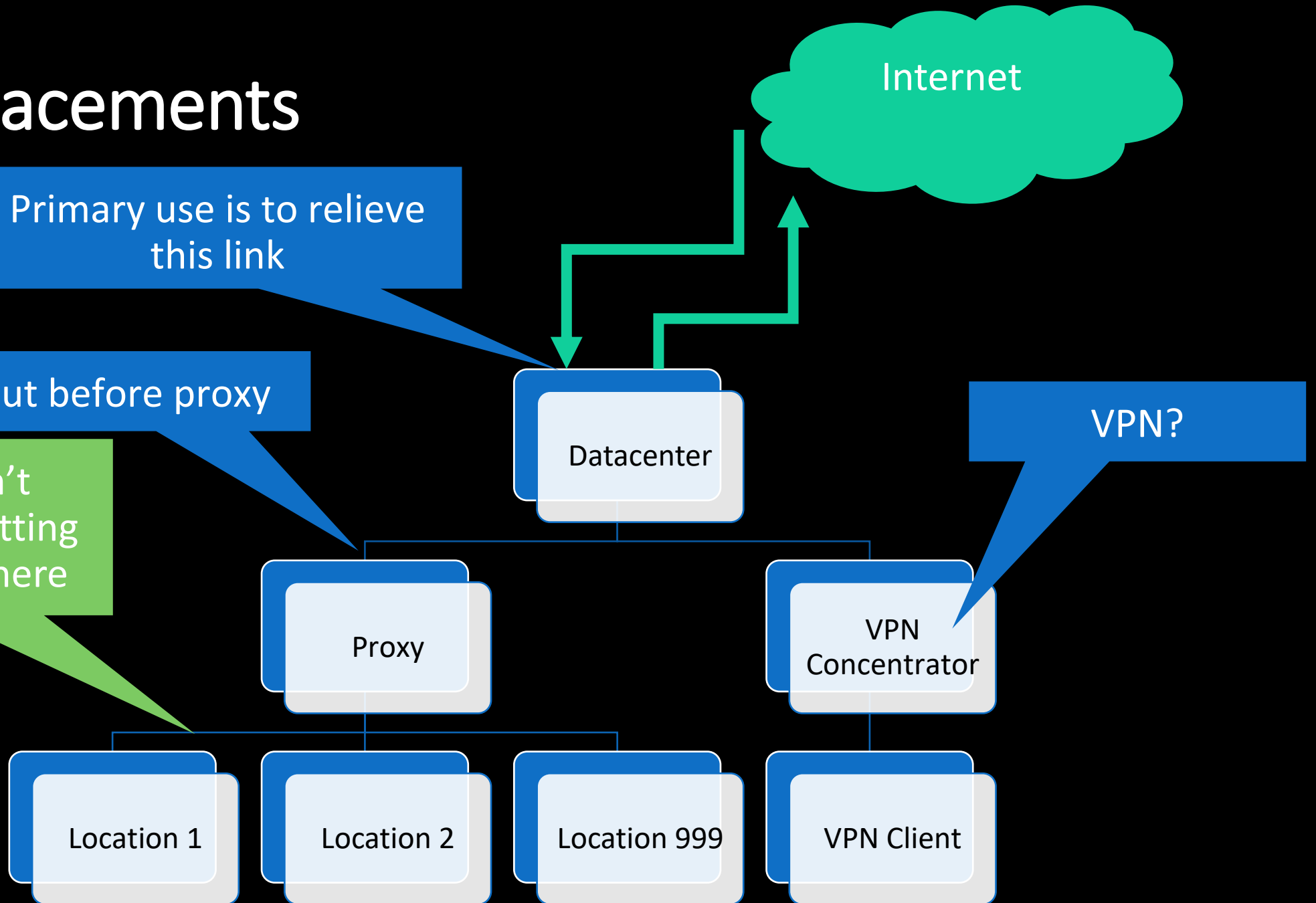
MCC placements

Primary use is to relieve this link

Put before proxy

Typically don't recommend putting MCC's everywhere

VPN?



MCC improvements and roadmap

Improvements

- ◆ BYO Cert importation for TLS configuration! Hallelujah!
- ◆ Auto discovery features – DNS-SD, other ways...
- ◆ Cache node ID to WufB reports
- ◆ Simplified deployment to Windows

Roadmap (No date)

- ◆ FQDNs for ISP-hosted MCCs
- ◆ Reporting move from WufB to Intune/Autopatch
- ◆ MCC in Azure government cloud
- ◆ Deployment integrations with...

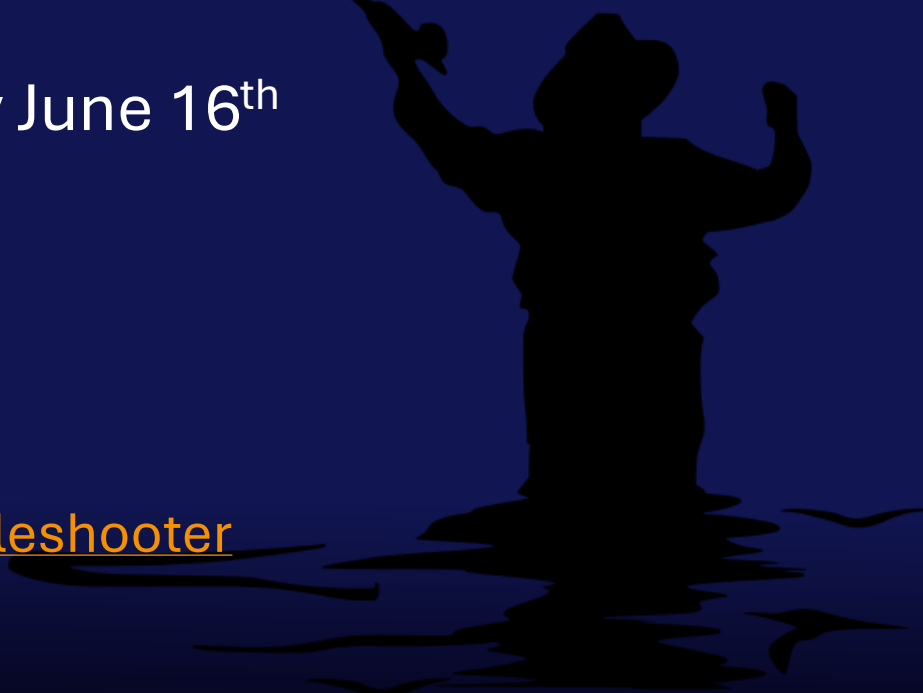
What to expect on the peering?

Windows Client	Minimum Windows version	HTTP Downloader	Peer to Peer	Microsoft Connected Cache
Windows Update (feature updates , quality updates , language packs , drivers)	Windows 10 1511, Windows 11	✓	✓	✓
Windows 10/11 UWP Store apps	Windows 10 1511, Windows 11	✓	✓	✓
Windows 11 Win32 Store apps	Windows 11	✓		
Windows Defender definition updates	Windows 10 1511, Windows 11	✓	✓	✓
Intune Win32 apps Learn more	Windows 10 1709, Windows 11	✓	✓ *	✓ *
Microsoft 365 apps and updates	Windows 10 1709, Windows 11	✓	✓ (excluding SAC Extended channel)	✓
Edge browser updates	Windows 10 1809, Windows 11	✓	✓	✓
Configuration Manager Express updates	Windows 10 1709 + Configuration Manager version 1711, Windows 11	✓	✓	✓
Dynamic updates	Windows 10 1903, Windows 11	✓	✓	✓
MDM Agent	Windows 11	✓		
Xbox Game Pass (PC)	Windows 10 1809, Windows 11	✓		✓
Windows Package Manager	Windows 10 1809, Windows 11	✓		
MSIX Installer	Windows 10 2004, Windows 11	✓		
Teams updates	Windows 10 2004, Windows 11	✓	✓ (starting with version 25122.1415.3698.68 12)	✓ (only over HTTPS)

<https://learn.microsoft.com/en-us/windows/deployment/do/waas-delivery-optimization>

So what's new in '26?

- ◆ Standalone Microsoft Connected Cache (MCC) – GA in 2025!!
 - ◆ (First announced at Ignite 2019, it only took 6 years...)
 - ◆ MCC + HTTPS
 - ◆ ConfigMgr + HTTPs with [Microsoft Connected Cache update for Microsoft Endpoint Configuration Manager version 2409, 2503 and 2509 - Configuration Manager | Microsoft Learn](#)
- ◆ Intune Win32 Apps delivered using Https by June 16th
 - ◆ Teams
- ◆ It's all about the slots!
- ◆ New features, new bugs?
- ◆ Troubleshooting tools
 - ◆ [PowerShell Gallery | DeliveryOptimizationTroubleshooter](#)



Https & Intune

- ♦ Starting on June 16th, 2026, or soon after...
 - ♦ Intune will enforce HTTPS content delivery for customers using Microsoft Connected Cache for Enterprise and Education.
- ♦ Companies using regular caching proxies will see influx of connections to the Internet!
- ♦ [How to enable HTTPS support for Microsoft Connected Cache for Enterprise and Education | Microsoft Community Hub](#)

Peering for Teams!

- ◆ Teams installations now peer, but ONLY over HTTPS.
- ◆ Remember to HTTPS enable your MCC's

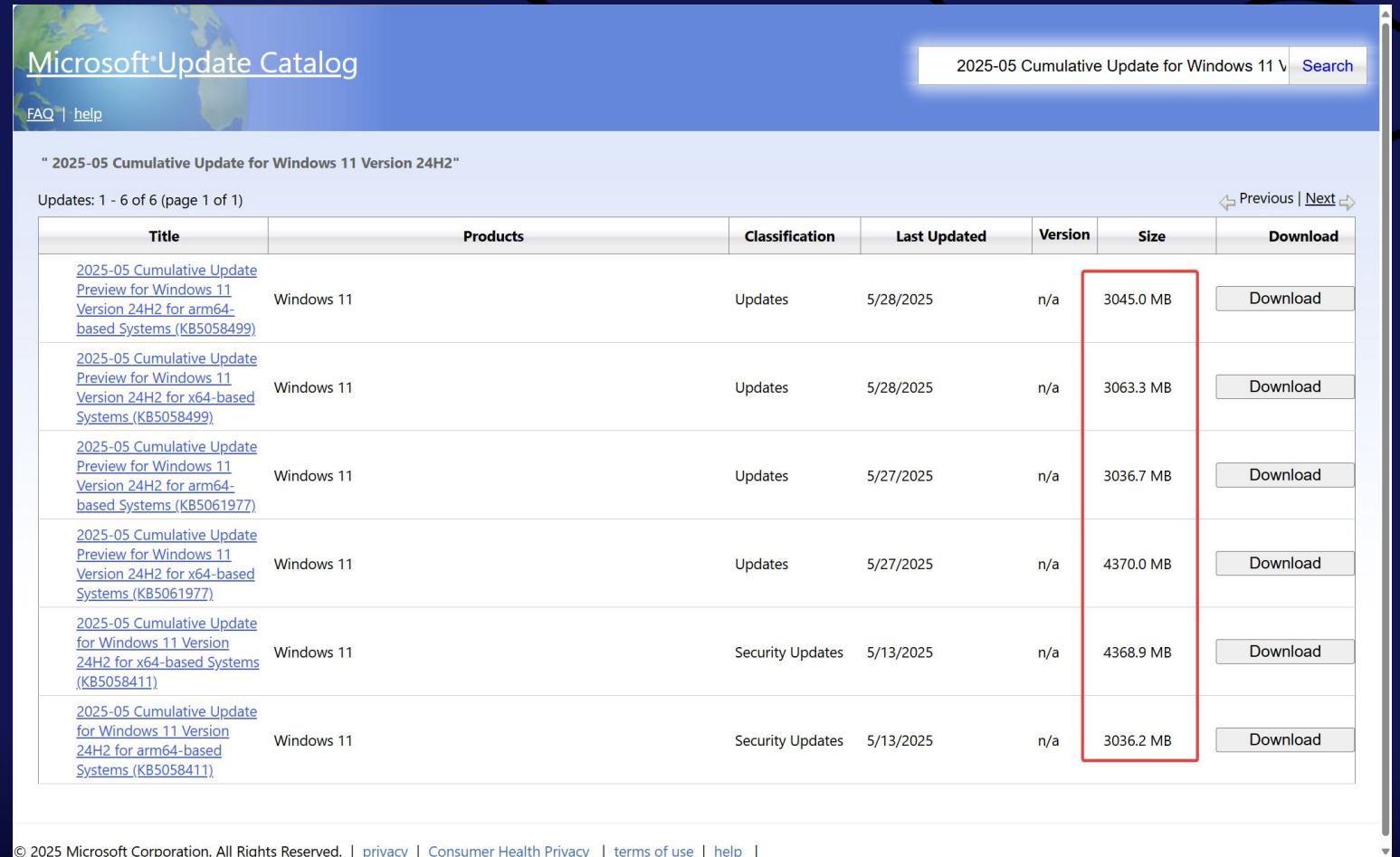
<https://learn.microsoft.com/en-us/windows/deployment/do/mcc-ent-https-windows>

That's a lot of DO traffic!

The security update for last May was **over 4 Gigabyte** in size!

June, July and August was each 3Gb

Takeaway: Even in good bandwidth location, make sure DO peering is working or that will be a lot going over your internet link.



Microsoft Update Catalog

2025-05 Cumulative Update for Windows 11 v Search

FAQ | help

" 2025-05 Cumulative Update for Windows 11 Version 24H2 "

Updates: 1 - 6 of 6 (page 1 of 1) Previous | Next

Title	Products	Classification	Last Updated	Version	Size	Download
2025-05 Cumulative Update Preview for Windows 11 Version 24H2 for arm64-based Systems (KB5058499)	Windows 11	Updates	5/28/2025	n/a	3045.0 MB	Download
2025-05 Cumulative Update Preview for Windows 11 Version 24H2 for x64-based Systems (KB5058499)	Windows 11	Updates	5/28/2025	n/a	3063.3 MB	Download
2025-05 Cumulative Update Preview for Windows 11 Version 24H2 for arm64-based Systems (KB5061977)	Windows 11	Updates	5/27/2025	n/a	3036.7 MB	Download
2025-05 Cumulative Update Preview for Windows 11 Version 24H2 for x64-based Systems (KB5061977)	Windows 11	Updates	5/27/2025	n/a	4370.0 MB	Download
2025-05 Cumulative Update for Windows 11 Version 24H2 for x64-based Systems (KB5058411)	Windows 11	Security Updates	5/13/2025	n/a	4368.9 MB	Download
2025-05 Cumulative Update for Windows 11 Version 24H2 for arm64-based Systems (KB5058411)	Windows 11	Security Updates	5/13/2025	n/a	3036.2 MB	Download

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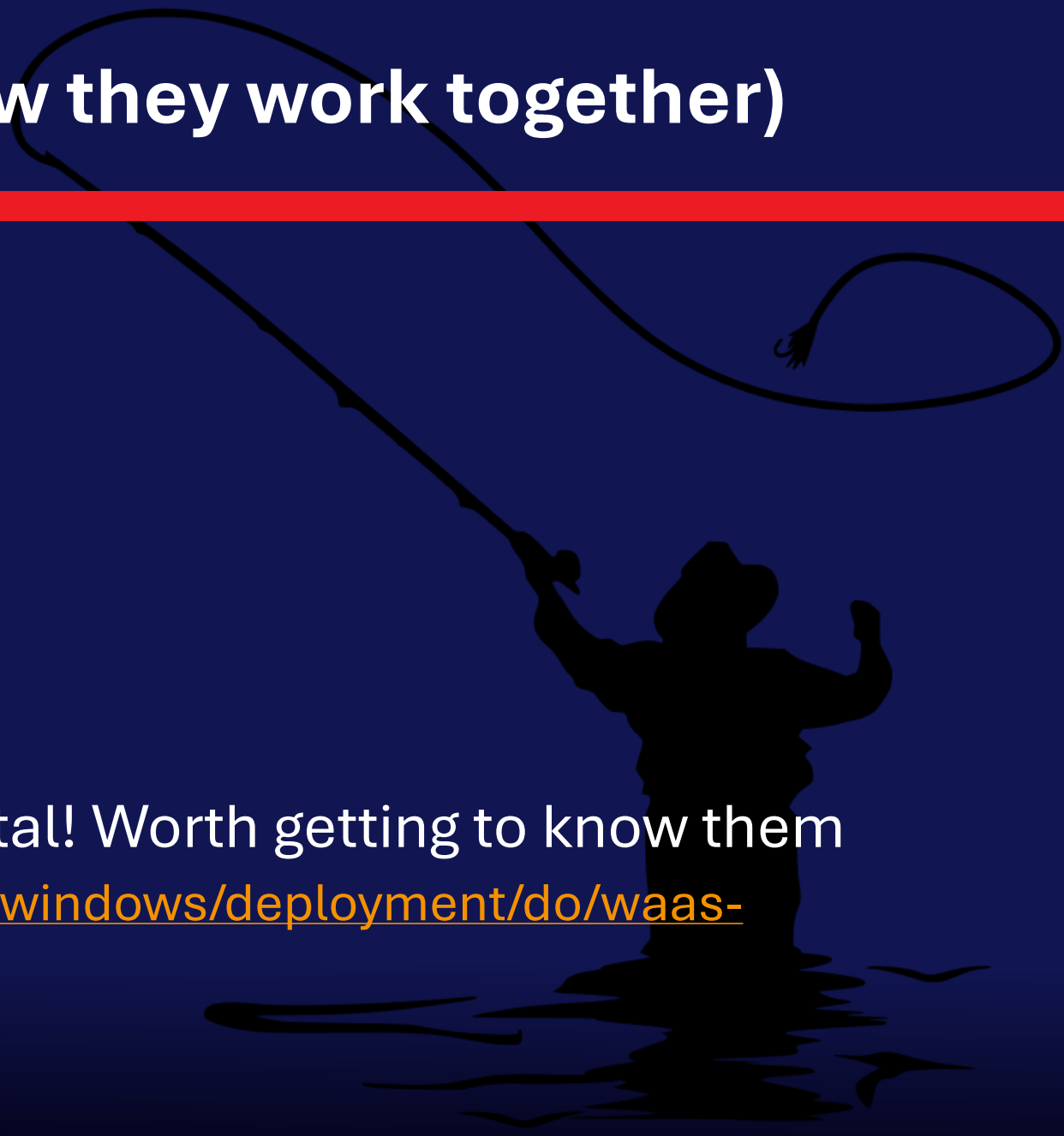
Optimizing DO

Aka 'Can I make it go faster? Or slower?'



Top 5 DO settings (and how they work together)

- ◆ Download Mode
 - ◆ Restrict Peer Selection (DNS-SD)
 - ◆ Background QoS
 - ◆ Delay
 - ◆ Cache Retention/file size
-
- ◆ There are more policies – 32 in total! Worth getting to know them
 - ◆ <https://learn.microsoft.com/en-us/windows/deployment/do/waas-delivery-optimization-reference>



Intune DO Starter Kit

Edit profile - 2Pint - Delivery Optimization

Settings catalog

[+ Add settings](#)

Delivery Optimization

[Remove category](#)

27 of 39 settings in this category are not configured

DO Min Background Qos *	64	
DO Absolute Max Cache Size *	30	
DO Allow VPN Peer Caching	Not allowed	
DO Delay Background Download From Http *	600	
DO Delay Foreground Download From Http *	60	
DO Download Mode	HTTP blended with peering behind the same NAT	
DO Max Cache Age *	0	
DO Min Battery Percentage Allowed To Upload *	40	
DO Min File Size To Cache *	1	
DO Min RAM Allowed To Peer *	2	
DO Monthly Upload Data Cap *	0	
DO Restrict Peer Selection By	Subnet mask	

Settings picker

Use commas "," among search terms to lookup settings by their keywords

delivery opt

Search

+ Add filter

Browse by category

Delivery Optimization

39 results in the "Delivery Optimization" category

Select all these settings

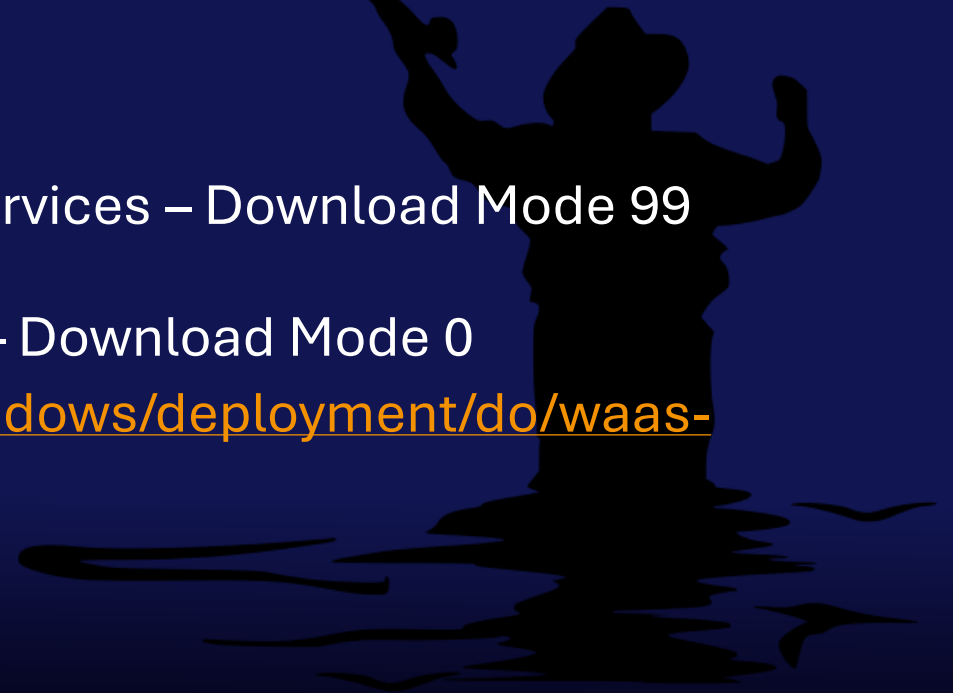
Setting name

☐	DO Max Foreground Download Bandwidth	
☐	DO Max Upload Bandwidth	
☒	DO Min Background Qos	
☒	DO Min Battery Percentage Allowed To Upload	
☐	DO Min Disk Size Allowed To Peer	
☒	DO Min File Size To Cache	
☒	DO Min RAM Allowed To Peer	
☐	DO Modify Cache Drive	
☒	DO Monthly Upload Data Cap	
☐	DO Percentage Max Background Bandwidth	
☐	DO Percentage Max Foreground Bandwidth	
☒	DO Restrict Peer Selection By	



Download Mode/Restrict Peer Selection

- ◆ Most Popular - 'Group' Modes 1 or 2 (and maybe 3?)
- ◆ 1 = Peering behind NAT 2 = peering between peering groups
- ◆ GroupID (GUID) is used to create peering groups
- ◆ GroupID can be set by
 - ◆ ConfigMgr
 - ◆ DHCP
 - ◆ DNS Suffix
 - ◆ AAD Tenant ID
- ◆ If you want DO to just go fast – bypass the Cloud Services – Download Mode 99
 - ◆ No peering or caching
- ◆ If peering is not working and you use MCC instead – Download Mode 0
- ◆ Learn More: <https://learn.microsoft.com/en-us/windows/deployment/do/waas-delivery-optimization-reference#download-mode>

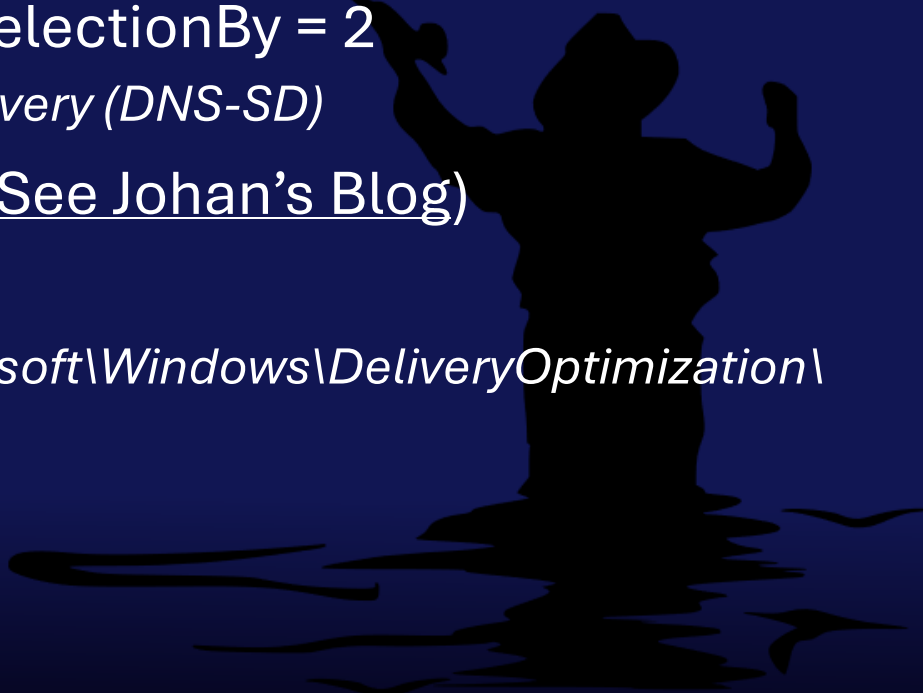


What is DNS-SD and is it any good?

- ◆ Service Discovery Protocol https://en.wikipedia.org/wiki/Multicast_DNS
- ◆ Uses multicast messages to find local peers (same subnet only unless using 2Pint 🍺)
 - ◆ Can be blocked by wi-fi Access Points (even when on the same subnet)
- ◆ IPv4 address 224.0.0.251 or IPv6 address ff02::fb
- ◆ Doesn't rely on the DO Cloud service (for peering) 🙌
- ◆ Useful for dealing with small files that are ignored by default (they all add up!)

What is DNS-SD and is it any good?

- ♦ With 'Group' Download Mode (1,2), DO cache now has 36 slots (which are rotated) for sharing content (was 4 before)
- ♦ DNS-SD has no restriction in W11 (W10 is still restricted to 4 slots!!) so ALL content peerable
- ♦ DNS-SD is enabled by setting DORestrictPeerSelectionBy = 2
 - ♦ 0 = None, 1 = Subnet mask, and 2 = Local Peer Discovery (DNS-SD)
- ♦ Can be configured using MDM Policy in Win11 ([See Johan's Blog](#))
- ♦ Win10 needs a registry poke (no policy)
 - ♦ `HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\DeliveryOptimization\DORestrictPeerSelectionBy` – set it to 2



Windows 11 Peer “slots” increased

2024
4 slots

2026
36 slots!!

Administrator: Windows PowerShell										FileId	Priority	Status	TotalBytesDownloaded	BytesFromLinkLocalPeers	BytesFromLanPeers	BytesFromCacheServer	PercentPeerCaching
>> select-object FileId,Priority,@[L='File (MB)';E='{"0:N2}" -f (\$_.FileSize / 1MB)'}],Status,DownloadMode,@[L='Downloaded (MB)';E='{"0:N2}" -f (\$_.TotalBytesDownloaded / 1MB)'}],@[L='Group Peers (MB)';E='{"0:N2}" -f (\$_.BytesFromGroupPeers / 1MB)'}],@[L='MCC (MB)';E='{"0:N2}" -f (\$_.BytesFromCacheServer / 1MB)'}],@[L='PeerPercent';E='{"0:N2}" -f (\$_.PercentPeerCaching)'] ft																	
FileId	Priority	File (MB)	Status	DownloadMode	Downloaded (MB)	LAN Peers (MB)	Group Peers (MB)	MCC (MB)	UploadedToPeers	de47fbb6f781466493029483a4a279cc74ce2ad	Background	Complete	2268900	2268900	2268900	0	100
64e0e2cb6d2e6645a6b238fb0707605091fee7	Background	177.63	Complete	Group	43.63	38.63	0.00	0.00	0.00	27045c5db6d6f3737e17d836fac59be4a2c4fbbad	Background	Caching	14999221	14999221	14999221	0	100
acb38d66466232b6ab1697b29d3561deb1f9fefdf	Background	71.27	Caching	Group	71.27	45.27	0.00	0.00	6.00	dfbda30b3df501217b5d43ad77d726058a9a6cb3	Background	Caching	40959020	0	0	40959020	0
CxRcUSHuK0Q0TuSy7NwLav1_Ms610U0UCS28DFktXkw8=	Background	165.85	Complete	Group	0.00	0.00	0.00	0.00	0.00	b9e9693b49a98254d193665484e20dde04d6b15c	Background	Caching	62837514	0	0	0	0
256739244b9479e5dc8f5e6258b1f0a1759fe00c	Background	70.77	Complete	Group	18.77	18.77	0.00	0.00	0.00	FkY5wwToGSRT-1Z_AxbGWF3Q0QR5WZUgA_gI0tSL1S8=	Background	Caching	1648184	0	0	1648184	0
bc1381bdf628911f6d90b96594a5912995aa5046	Background	424.29	Complete	Group	0.00	0.00	0.00	0.00	0.00	96020d1b6450a6b7dcbec4581b3867c25173d993	Background	Caching	2353440	2353440	2353440	0	100
877ae0c3c54af53cc6853b51702f62ba172c8493	Background	135.69	Complete	Group	0.00	0.00	0.00	0.00	0.00	202d5bdecfa198fa18d26118d878ca3b84573b04	Background	Caching	2336609	2336609	2336609	0	100
4e5850f5d66bd8e3b51955519340bf282c1dee88	Background	162.63	Complete	Group	81.63	81.63	0.00	0.00	0.00	c9a2e1383d09d6dea080e752a0e5550858148e7b	Background	Caching	42224606	0	0	0	0
a088c1c7e80d56ae6f520f63d7745a16a87b64ae	Background	177.92	Complete	Group	42.92	0.00	0.00	0.00	0.00	a5880ce7e7e63544c2a09ebf2640087f9d12d3f	Background	Caching	1522112	0	0	0	0
f786417fbcec3039e78101d55d5632f44b7d35b	Background	72.39	Complete	Group	72.39	0.00	0.00	0.00	72.39	2921cfcf97acd51d4e9efae071570a4b3f8589796	Background	Caching	1276328	0	0	0	0
9d1fcfaa5926fceb63183a17c632d837fed36fd1	Background	161.54	Complete	Group	8.54	0.00	0.00	0.00	0.00	06d6c2e0f01aa65b11a48a49d06b4c7f1e8a3fe	Background	Caching	11399002	0	0	0	0
dcb0e0a889a235d81baefc-b281fa77303281753	Background	70.79	Complete	Group	0.00	0.00	0.00	0.00	0.00	259b3ca8c86aab18d537852d6818c-fb27237e29	Background	Caching	23127681	18874368	18874368	0 ... 4272486723	0
93d628e0fd1079e389a902532ae45300b677f47c	Background	424.29	Complete	Group	58.29	58.29	0.00	0.00	0.00	dbda1bbdcdb81ee3fc0361d5766f1ec54da4ce2a	Background	Caching	2269954	0	0	2269954	0
as992f6ceb0dddf1331f7045e8a35891ad6fb308c	Background	70.28	Complete	Group	13.28	0.00	0.00	0.00	0.00	f7d9f7fc3fe143b5ced02ce5f9baee3d2d80800dd	Background	Caching	55400585	55400585	55400585	0	100
85277b1a5d99318b6ba8cd75ddfd4807f9bce6e	Background	252.48	Complete	Group	12.48	0.00	0.00	0.00	10.73	ace81cfff6a2fb73a7b1ec2504dae8499a5466dccc	Background	Caching	53017197	0	0	53017197	0
1190e656a405bfbf7e43602ed079aef7354615ea	Background	304.13	Complete	Group	25.13	125.13	0.00	0.00	0.00	da900f8f9c34402e47e848d24d878967f5401bfff	Background	Caching	52576048	49283072	49283072	3292976 ... 7373066914	0
0e6b955f9563b68a1f4a8de0c93dd49f47d282869	Background	53.82	Complete	Group	22.82	0.00	0.00	0.00	0.00	27ee385ce691c23beed10d6cf667e937012ecac	Background	Caching	1505704	0	0	1505704	0
04d397dd9159f3ab941581272f847a5846750ad3	Background	290.10	Complete	Group	110.10	0.00	0.00	0.00	0.00	ad5d43e3c66ea355a3c2f9e9bfb6cccf55c199109	Background	Caching	20175201	16777216	16777216	0 ... 6151335494	0
34926b4a8016dbdb7ea821f79f656ab81d656a03	Background	121.59	Complete	Group	40.59	0.00	0.00	0.00	0.00	5302a428dac2108fb33c136ddfe7fe118a1f432	Background	Caching	54471044	0	0	8333700	0
b09453ddb56f66d7ecf0f9d0b81970acbe770abf	Background	468.02	Complete	Group	3.02	3.02	0.00	0.00	0.00	bb35b4bb5cbfd10e0537b397dd2fa23e0c157167	Background	Caching	43199644	38797312	38797312	4402332 ... 3326880194	0
2dec4ae7406a8a9948f3cc3ca06f4c68ca0885be	Background	300.49	Complete	Group	124.49	0.00	0.00	0.00	0.00	22e3b14588240d17b9d2fe9424ebdd50893952	Background	Caching	5791607	0	0	0	0
195f82e048d7bf91f261c077176127ecc0019f4b	Background	53.86	Complete	Group	0.00	0.00	0.00	0.00	0.00	92d7048b28de73ef74220196ccc37dd3075278cc	Background	Caching	75773826	0	0	0	0
77949b9f8a55c4f4358ac3d8e82c00fc0bbfb73	Background	155.47	Complete	Group	114.47	0.00	0.00	0.00	0.00	ketCaqUUs395xcCmg5PAFjAXX1TLd0G620yka1AMvPg=	Background	Caching	64186448	0	0	1048576	0
cd369cfc3ecd2e67c715dc28e563ca7ac15157f9	Background	248.80	Caching	Group	301.48	0.00	0.00	0.00	0.00	e23d9f4970656e0bf6d4aa2b0eac8d23c509215	Background	Caching	53617528	0	0	53617528	0
05bbdd2c09c300db2f12ef1a1b9b94b180fba6da	Background	83.75	Complete	Group	29.75	0.00	0.00	0.00	0.00	f1005f2f2b0a9cbec3725ac19c9a0f6dfccff0a	Background	Caching	2558434	2558434	2558434	0	100
5801c05d4de4a91bd3afbf01e78980d4344285061	Background	70.71	Complete	Group	20.71	0.00	0.00	0.00	0.00	60a9910b62c0035e74b11ce09f005699527b69e	Background	Caching	75159600	0	0	75159600	0
9a358199c4c07502f01c0b20348853105787125	Background	92.62	Complete	Group	31.62	0.00	0.00	0.00	0.00	1f99a50a397f6ea6720333de91929574de03081	Background	Caching	384827392	308041312	308281344	1048576 ... 9918256131	0
903c4701a013c7f46e24ce2b2f0eb9f04bf02d2f6	Background	50.53	Complete	Group	10.53	0.00	0.00	0.00	0.00	549595164f395705f50b19c5fe08c1842fb7fba3	Background	Caching	25955654	25955654	25955654	0	100
ee105a89f8ac44c060488bf57e2fbbfbf5817e92	Background	178.75	Complete	Group	45.75	0.00	0.00	0.00	45.75	a7dfbb8412682a3a7d3263f082ba70df7eac2c2ab9	Background	Caching	2448302	0	0	2448302	0
1510ba4c6a210b89d1f751b9db08c0450b123cd3	Background	699.60	Complete	Group	699.60	698.97	0.00	2,680.45	0.00	5cd6feed29e15a49d85e014e063e21ec886f3ea	Background	Caching	20637400	16777216	16777216	0 ... 8816340619	0
639d45d1ae7f9cf2570027f52e31c03ce20380b	Background	730.92	Complete	Group	730.92	0.00	0.00	730.92	0.00	525b3bca1a2f87f45bba4cf3616562a937466c4	Background	Caching	17097108	17097108	17097108	0	100
c396663579e9f15702a4a89f974195591ab8461ab	Background	153.18	Caching	Group	148.18	0.00	0.00	0.00	0.00	v5IdMbIDPG-blEwM2PzG7FCRMHUNAiuXCIvFg8Hj9U=	Background	Caching	13860944	0	0	1278032	0
ac9c1522c4a577bdf052c5ff5900a0514d71f72a	Background	156.57	Complete	Group	0.00	0.00	0.00	0.00	0.00	d6f10ea6440c8d9fe41977573b1e4c8943ba91e	Background	Caching	1261026	1261026	1261026	0	100
e94bf8b997f501f521be598c21d5604aa730f73	Background	793.86	Complete	Group	47.86	0.00	0.00	0.00	0.00	18dc3748ad96605fca389a80287b64a7c1e838c8	Background	Caching	16154110	0	0	2522622	0
YqVy3Ch-YpNp_QpH_CB2bZcNorR_FnhWkybGX7HKc=	Background	165.74	Complete	Group	165.74	145.74	0.00	0.00	13.63	b48b9b2ab71d68f49b435f51faf834a382a13b4	Background	Caching	230287566	0	0	230287566	0
587407dfd277efa0c316e02802343164a0d1d5d	Background	178.03	Caching	Group	47.03	0.00	0.00	0.00	0.00	ea179cd51ec1b49f4c4ae02a6843a6a31d017292	Background	Caching	2151652	0	0	1485232	0
034e1d088b1f8a2f0d23da13f53815611f429ba	Background	792.33	Complete	Group	97.33	93.58	0.00	0.00	0.00	e653ba5814359f723eb3e473458d503591601415	Background	Caching	0	0	0	0	0
80817f5bf6ed6e40e16023d1fd8ddafba52374	Background	424.29	Complete	Group	56.29	56.29	0.00	0.00	0.00								
30794c19f3e7e4fc23e6ac87518a62bc89fb83a7	Background	99.77	Complete	Group	18.77	18.77	0.00	0.00	0.00								
381db67c0b075e981714bb62c2b91091f68d5d7	Background	151.26	Complete	Group	91.26	0.00	0.00	0.00	0.00								
953449672073f8fb99bad4cc6d5d7849b9c83e8	Background	509.03	Complete	Group	307.00	284.00	0.00	23.00	0.00								
64b11788eb6bc1425010f420fcec34474246e719	Background	70.34	Complete	Group	18.34	18.34	0.00	0.00	0.00								
1db8c5aacda76af1e3768d5405198a51d08b5b28	Background	70.86	Complete	Group	70.86	42.86	0.00	0.00	19.88								

PS C:\Users\Administrator> (Get-DeliveryOptimizationStatus).FileId.count

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Activate Windows

Go to Settings to activate Windows

MMS



DNS-SD and Windows 11

Windows 11 enables sharing for all content pieces when DNS-SD is enabled

FileId	Priority	Status	TotalBytesDownloaded	BytesFromLinkLocalPeers	BytesFromLanPeers	BytesFromCacheServer	PercentPeerCaching	FileId	Priority	Status	TotalBytesDownloaded	BytesFromLinkLocalPeers	BytesFromLanPeers	BytesFromCacheServer	PercentPeerCaching
de47fbfb6f781466493029483a4a279cc74ce2ad	Background	Complete	2268900	2268900	2268900	0	100	de47fbfb6f781466493029483a4a279cc74ce2ad	Background	Caching	2268900	2268900	2268900	0	100
27045cd6d6d6f3737e17d836fac59be4a2c4fbbad	Background	Caching	14999221	14999221	14999221	0	100	27045cd6d6d6f3737e17d836fac59be4a2c4fbbad	Background	Caching	14999221	14999221	14999221	0	100
dfbda30b3df501217b5d43ad77d726058a9a6cb3	Background	Caching	40959020	0	0	40959020	0	dfbda30b3df501217b5d43ad77d726058a9a6cb3	Background	Caching	40959020	0	0	40959020	0
b9e9693b49a98254d193665484e20dde04d6b15c	Background	Caching	62837514	0	0	0	0	b9e9693b49a98254d193665484e20dde04d6b15c	Background	Caching	62837514	0	0	0	0
FKY5mwToG5Rt-1Z_AxbGWF30Q0R5WZUgA_g10t5LiS8=	Background	Caching	1648184	0	1648184	0	0	FKY5mwToG5Rt-1Z_AxbGWF30Q0R5WZUgA_g10t5LiS8=	Background	Caching	1648184	0	1648184	0	0
96020d1b6450a6b7dcb4581bd3867c25173d993	Background	Caching	2353440	2353440	2353440	0	100	96020d1b6450a6b7dcb4581bd3867c25173d993	Background	Caching	2353440	2353440	2353440	0	100
202d5bdecfa190fa184026118d878ca3b84573b04	Background	Caching	2336609	2336609	2336609	0	100	202d5bdecfa190fa184026118d878ca3b84573b04	Background	Caching	2336609	2336609	2336609	0	100
c9a2e1383d096dea800e752a0e5558958140e7b	Background	Caching	42224606	0	0	42224606	0	c9a2e1383d096dea800e752a0e5558958140e7b	Background	Caching	42224606	0	0	42224606	0
a5880ce7e7f63514c2a09ebf26400a7f0421d3f	Background	Caching	1521212	0	0	0	0	a5880ce7e7f63514c2a09ebf26400a7f0421d3f	Background	Caching	1521212	0	0	0	0
2921cff97acd51d1e9effe071570e4b3f8589796	Background	Caching	1276328	0	0	0	0	2921cff97acd51d1e9effe071570e4b3f8589796	Background	Caching	1276328	0	0	0	0
06d6c2ce0f01aaa65b11a48a49d06b4c7f1e8e3fe	Background	Caching	11399022	0	0	0	0	06d6c2ce0f01aaa65b11a48a49d06b4c7f1e8e3fe	Background	Caching	11399022	0	0	0	0
259b3ca8c06aabd18d537852d6818cfb27237e29	Background	Caching	23127681	18874368	18874368	0	...	259b3ca8c06aabd18d537852d6818cfb27237e29	Background	Caching	23127681	18874368	18874368	0	...
dbda1bbdcdb81ee3fc0361d5766f1ec54da4ce2a	Background	Caching	2269954	0	2269954	0	100	dbda1bbdcdb81ee3fc0361d5766f1ec54da4ce2a	Background	Caching	2269954	0	2269954	0	100
f7d9f7c3fe143b5ced02ce5f9baee3d28d0800dd	Background	Caching	55400585	55400585	55400585	0	100	f7d9f7c3fe143b5ced02ce5f9baee3d28d0800dd	Background	Caching	55400585	55400585	55400585	0	100
ace81cff6a2fb73a7b1ec2504dae8499a5466dcc	Background	Caching	53017197	0	0	53017197	0	ace81cff6a2fb73a7b1ec2504dae8499a5466dcc	Background	Caching	53017197	0	0	53017197	0
da900f9f9c34042e47e848d24d878967f5041bff	Background	Caching	52576048	49283072	49283072	0	...	da900f9f9c34042e47e848d24d878967f5041bff	Background	Caching	52576048	49283072	49283072	0	...
27ee385ce691c23beed10d6cf667e937012ecac	Background	Caching	1505704	0	0	0	0	27ee385ce691c23beed10d6cf667e937012ecac	Background	Caching	1505704	0	0	0	0
as4d43e3c66ea355a3c2f9e9bfb6ccecf55c199109	Background	Caching	20175201	16777216	16777216	0	...	as4d43e3c66ea355a3c2f9e9bfb6ccecf55c199109	Background	Caching	20175201	16777216	16777216	0	...
5302a428dac2108fb33c3136ddfe7fe118a1f432	Background	Caching	54471044	0	0	54471044	0	5302a428dac2108fb33c3136ddfe7fe118a1f432	Background	Caching	54471044	0	0	54471044	0
bb35b4bb5cbfd10e0537b397dd2fa23e0c157167	Background	Caching	43919644	38797312	38797312	0	...	bb35b4bb5cbfd10e0537b397dd2fa23e0c157167	Background	Caching	43919644	38797312	38797312	0	...
22e3b14588244d017b9d2fe94242ebdd50893952	Background	Caching	5791607	0	0	0	0	22e3b14588244d017b9d2fe94242ebdd50893952	Background	Caching	5791607	0	0	0	0
92d7048b28de73ef74220106ecc37dd3075278cc	Background	Caching	75773826	0	0	0	0	92d7048b28de73ef74220106ecc37dd3075278cc	Background	Caching	75773826	0	0	0	0
kEtCAqUUsJ95xCGmg5PAFjaXX1TLd0Gb20yka1AMVpg=	Background	Caching	64186448	0	0	64186448	0	kEtCAqUUsJ95xCGmg5PAFjaXX1TLd0Gb20yka1AMVpg=	Background	Caching	64186448	0	0	64186448	0
e23d9349700566ce0fbfd4aa2b8ca8dc23c509315	Background	Caching	53617584	0	0	53617584	0	e23d9349700566ce0fbfd4aa2b8ca8dc23c509315	Background	Caching	53617584	0	0	53617584	0
f1005f2f2b0a98cbec3725ac19c9e0fa4fccff0a	Background	Caching	2558434	2558434	2558434	0	100	f1005f2f2b0a98cbec3725ac19c9e0fa4fccff0a	Background	Caching	2558434	2558434	2558434	0	100
60a9940b62c00835e74b11ce9ff005699527b69e	Background	Caching	75459600	0	0	0	0	60a9940b62c00835e74b11ce9ff005699527b69e	Background	Caching	75459600	0	0	0	0
1f99a50da97f6ea6720333de91929574de003081	Background	Caching	380827392	300941312	300941312	0	...	1f99a50da97f6ea6720333de91929574de003081	Background	Caching	380827392	300941312	300941312	0	...
5d9595164f395705f50b19c5fe0c1842fb7fba3	Background	Caching	25955654	25955654	25955654	0	100	5d9595164f395705f50b19c5fe0c1842fb7fba3	Background	Caching	25955654	25955654	25955654	0	100
a7dfb8412682a3a7d3263fd02ba70d7eac2c2ab9	Background	Caching	2448302	0	0	0	0	a7dfb8412682a3a7d3263fd02ba70d7eac2c2ab9	Background	Caching	2448302	0	0	0	0
5cd6feed29e15a49d845e014e063e21ec886f3ea	Background	Caching	20163740	16777216	16777216	0	...	5cd6feed29e15a49d845e014e063e21ec886f3ea	Background	Caching	20163740	16777216	16777216	0	...
525b3bca1a2f87f45bba4cf3616562a937466c4	Background	Caching	17097108	17097108	17097108	0	100	525b3bca1a2f87f45bba4cf3616562a937466c4	Background	Caching	17097108	17097108	17097108	0	100
vS1dMbIDPc-b1EuW2PrG7FCRMHUNA1ZuXCVg9hJ9U=	Background	Caching	13860904	0	0	1278032	0	vS1dMbIDPc-b1EuW2PrG7FCRMHUNA1ZuXCVg9hJ9U=	Background	Caching	13860904	0	0	1278032	0
d6f10ea6400e8d9f4e1977573be14dc8943ba91e	Background	Caching	1261026	1261026	1261026	0	100	d6f10ea6400e8d9f4e1977573be14dc8943ba91e	Background	Caching	1261026	1261026	1261026	0	100
18dc3748a96605fca389a80287b64a7c1e038c8	Background	Caching	16154110	0	0	0	0	18dc3748a96605fca389a80287b64a7c1e038c8	Background	Caching	16154110	0	0	0	0
b48b9b2ab71d68f49b435f51fa834a8382a13b4	Background	Caching	230287566	0	0	0	0	b48b9b2ab71d68f49b435f51fa834a8382a13b4	Background	Caching	230287566	0	0	0	0
ea179cd51ec1b4fc4a0e42a60434a6a31d017292	Background	Caching	2151652	0	0	0	0	ea179cd51ec1b4fc4a0e42a60434a6a31d017292	Background	Caching	2151652	0	0	0	0
e653ba5814359f723eb3e473458d503591601415	Background	Caching	1485232	0	0	0	0	e653ba5814359f723eb3e473458d503591601415	Background	Caching	1485232	0	0	0	0

PS C:\Users\Administrator> (Get-DeliveryOptimizationStatus).FileId.Count

37

Activate Windows
Go to Settings to activate Windows.

PS C:\Users\Administrator> (Get-DeliveryOptimizationStatus).FileId.Count

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Activate Windows
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DNS-SD and Windows 10

Windows 10 with DNS-SD enabled

Administrator: Windows PowerShell

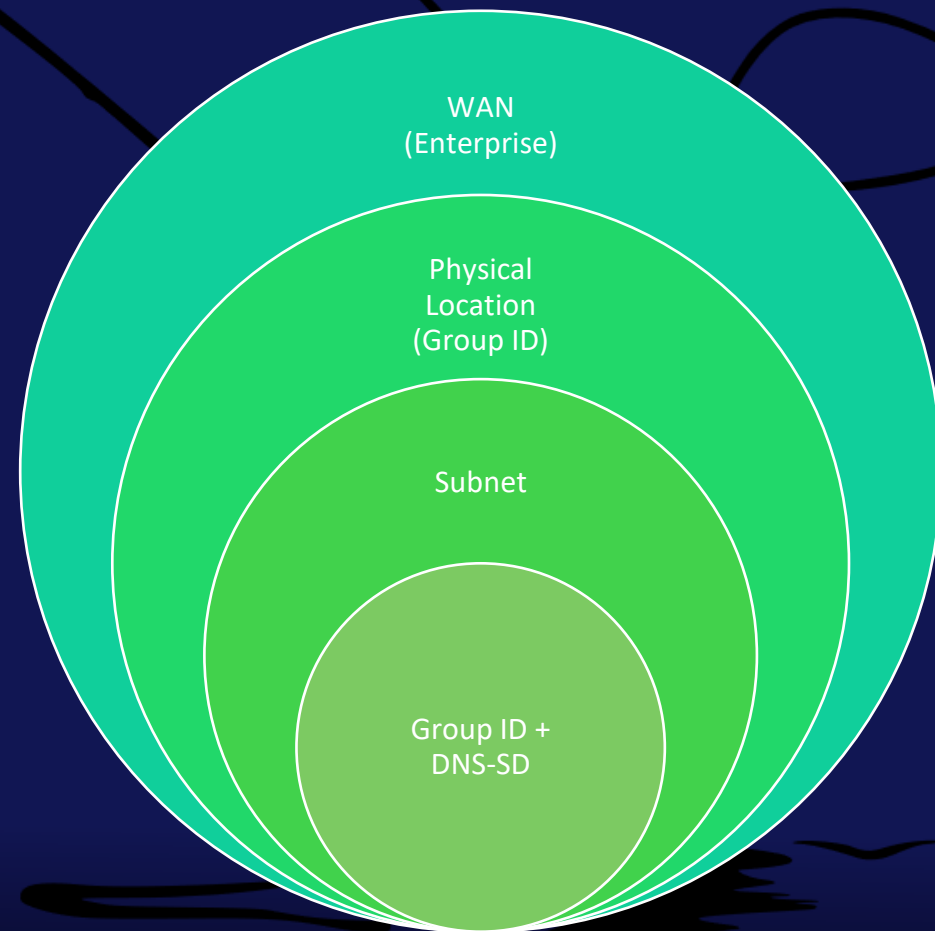
```
PS C:\windows\system32> Get-DeliveryOptimizationStatus | select-Object fileid,priority,status,totalbytesdownloaded, BytesFromLinkLocalPeers,bytesfromlanpeers,bytesfromcache
```

FileId	Priority	Status	TotalBytesDownloaded	BytesFromLinkLocalPeers	BytesFromLanPeers	BytesFromCacheServer	PercentPeerCaching
34926b4a8016dbbd7ea821f79f656ab81d6564ab	Background	Complete	42563350		42563350	0	100
aecb5c724f0b2a8cfa1730f02cfaf89ced871e55	Background	Complete	98988085	0		0	0
7f82e42f560636ebe9bfa9f577ae1485fd85ab07	Background	Complete	100737740	0		0	0
5ea4987246beeb7a0397acfb38cdcb9fe6f09276	Background	Complete	12338776	0		0	0
33d628e9fd1079e389a992532ae45308b677f47c	Background	Complete	41198713	0		0	0
YqVy3Ch-Yp9Np_QpH_CB2bcNorR____FnhWkybGX7HKc=	Background	Complete	173792864		173792864	0	100
34b50bac756cd7325205d3d9bclae8c093b95d2	Background	Complete	55012285	0		0	0
Fcc5831e5c71ec8ef93783847b70411e2635cecd	Background	Complete	13264045	0		0	0
vyENji9o3YtmZL97iB0RC0o0YPA2MacCB5WjLpd4v6k=	Background	Complete	40656456	0		0	0
2f520e0a0b2f037152b7758346e2d0c9d6b8ab28	Background	Complete	56977960	0		0	0
e878783adb6a33e731963530c1d372c636a62f88	Background	Caching	26582456		25533880	0	96.055383294907
c7e7aef760484a944b1fa2cfa26f58549cfef2bd	Background	Complete	33919566	0		0	0
c12e1a02e8ae7b45217c8116c4d859b8d869b1ad	Background	Paused	0	0		0	0
5N6jdmw446aL_19Xf8eCv2ZwxRJ0CKpKjRvhV4g04sQ=	Background	Complete	40643136	0		0	0
50233a1ab2c4d2c167b60064fb9d3fcf9de50ad5	Background	Caching	131840436	0		0	0
1510b4c86a210b89d1f751b9deb8c0458b123cd3	Background	Complete	733580610		612994370	0	83.5619646489838
5bbb6c6c4f04886280a275d3dd7e6e0458025255	Background	Complete	7673142	0		0	0
7f6f7642033c9b925face37d64a1ba9f7199e2df	Background	Complete	45911951		23891855	0	52.0384224142424
a554ce2a6750a1eeb4bccd39bdd506aa044822c	Background	Complete	104171940		104171940	0	100
4612378a6383202b033f49becaaf2448dbd715fc	Background	Complete	19813589	0		0	0
54b11788eb6bc1425010f420fcce34474246e719	Background	Complete	19227358		19227358	0	100
3f3ca31a0504fa081a331133ada6bf672820da2d	Background	Complete	22312479	0		0	0
1dad006505b6fa37c5872de45febb6f31efa6e0a	Background	Complete	22577995		13140811	0	58.2018509615225
54116967b608c9d7cf29687819b6234d0099ab52	Background	Complete	41305582	0		0	0
539d45d1ae7f9cf2578027f52e316c03ce2038b0	Background	Complete	560901590	0		0	0
35bbd2c09c300db42f12ef1a1b9b94b180fba6da	Background	Complete	31199505		7340032	0	23.5261168406358
381db67c0b0758e981714b62c2b91091f68d5d7	Background	Complete	95692159	0		0	0
59e6f902d2f7484117b62a007e88590b22c324ec	Background	Complete	20061092		19012516	0	94.7730861311039
1e466de692ea606bad3e59a1e7c69b75de77f7c8	Background	Complete	66636715	0		0	0
3195bfe441d7286d6586ea5ff7c8e8c68f52531c	Background	Complete	1235427	0		0	0
0792e28a6ef2612086670d3152b757ac29c9ac8f	Background	Complete	11940517		406181	0	3.40170362807574
bc1381bdf628911f6d90b96594a5912995aa5046	Background	Complete	61126129	0		0	0
377ae0c3c54af53cc6853b51702f62ba172c8493	Background	Caching	142276088		1048576	0	0.737000865528436
983c4701a613cf746e24ce2b2fa6bf94bfb2d2f6	Background	Complete	11041382	0		0	0
155612b01e9f1ff62c210160757e6cc180385304	Background	Caching	158234538	0		0	0
77949b9f8a55c4f4358ac3d8e82c00fcb0fbb473	Background	Complete	120030166	0		0	0
38f34ee85164c0fd0295d51bc5c6184623c66a6f	Background	Complete	19852172		19852172	0	100
3a358199c4c97562fb1c6b229348853105787125	Background	Complete	35251329		1696897	0	4.8137107114458
85277b1a5d99318b6bda0cd75ddfd4807f9bc6e6	Background	Complete	50832297		3145728	0	6.18844354013748



DNS-SD Limiting

- ◆ DNS-SD is always limited to subnet (broadcast)
- ◆ DO will register the peer via its client IP + the file hash + Group ID if set
- ◆ If using Group ID it will be a limiting factor. It is Group ID **AND** DNS-SD (not **OR**).
 - ◆ *“If Group mode is configured, Delivery Optimization connects to locally discovered peers that are also part of the same Group (have the same Group ID) and prevents devices that aren't using the same Group ID from participating.”*
- ◆ If you have multiple small networks in a single physical location, (one for each floor), consider to make one big one instead to improve DO peering. (or not ;-))
- ◆ Beware CISCO storm control



DNS-SD Pros and Cons

◆ PROS:

- ◆ When working, very efficient. Doesn't need to contact DO backend services.

◆ CONS:

- ◆ Uses Multicast (including replies). Multicast announcement every 5 mins.
- ◆ Effectively disables intra subnet peering.
- ◆ If not working – zero P2P! (no fallback to Group mode)

Takeaway: Only use DNS-SD where appropriate!

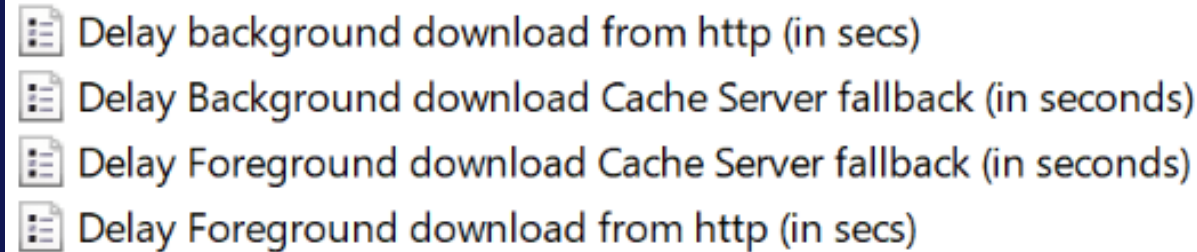
Background QoS

- ◆ Somewhat counter-intuitive?
- ◆ Dictates the **minimum** download speed from CDN
 - ◆ If too high – no P2P
 - ◆ Default is 2500KB/s (20MB/s)
- ◆ The lower this value is, the more content will be sourced using peers on the network rather than HTTP sources.
- ◆ **Takeaway: If you want to encourage more P2P lower this value from the default. If you just want to go fast, set this value higher (KB/s)**



Delay all the things

- ◆ Designed to improve peering and Connected Cache download probability
- ◆ Delays the DL while DO finds peers with content. Backend DO services might be busy
- ◆ Don't delay FG too much – users will not thank you!



- ◆ Delay background download from http (in secs)
- ◆ Delay Background download Cache Server fallback (in seconds)
- ◆ Delay Foreground download Cache Server fallback (in seconds)
- ◆ Delay Foreground download from http (in secs)

- ◆ Delay from http always applies first (to get content from peers if possible)
- ◆ **Takeaway: Use these if you want to improve peering, but don't set the values too high**

Cache retention/Minimum filesize

- ◆ Uses Max Cache Age + Minimum Peer Caching Content File Size policies
- ◆ If using Group Download Modes (1,2) setting a high MaxCacheAge/Small filesize can be counter productive
 - ◆ Remember the 'slot rule?' – Keeping content too long can lead to 'cache clutter'
 - ◆ Too many small files increases chance of them filling the 36 slots
 - ◆ **Takeaway: Cache retention no higher than 30 days (depends on content)**
 - ◆ **Takeaway: Minimum file size no lower than 10MB**
- ◆ If using DNS-SD, you can keep more content for longer
 - ◆ No slot rule – all content peerable (only on W11)
 - ◆ Even small files can peer
 - ◆ **Takeaway: If maximizing P2P, keep content for longer**
 - ◆ **Takeaway: Set Minimum file size to 1MB to hoover up the small stuff**
 - ◆ Bear in mind cache size

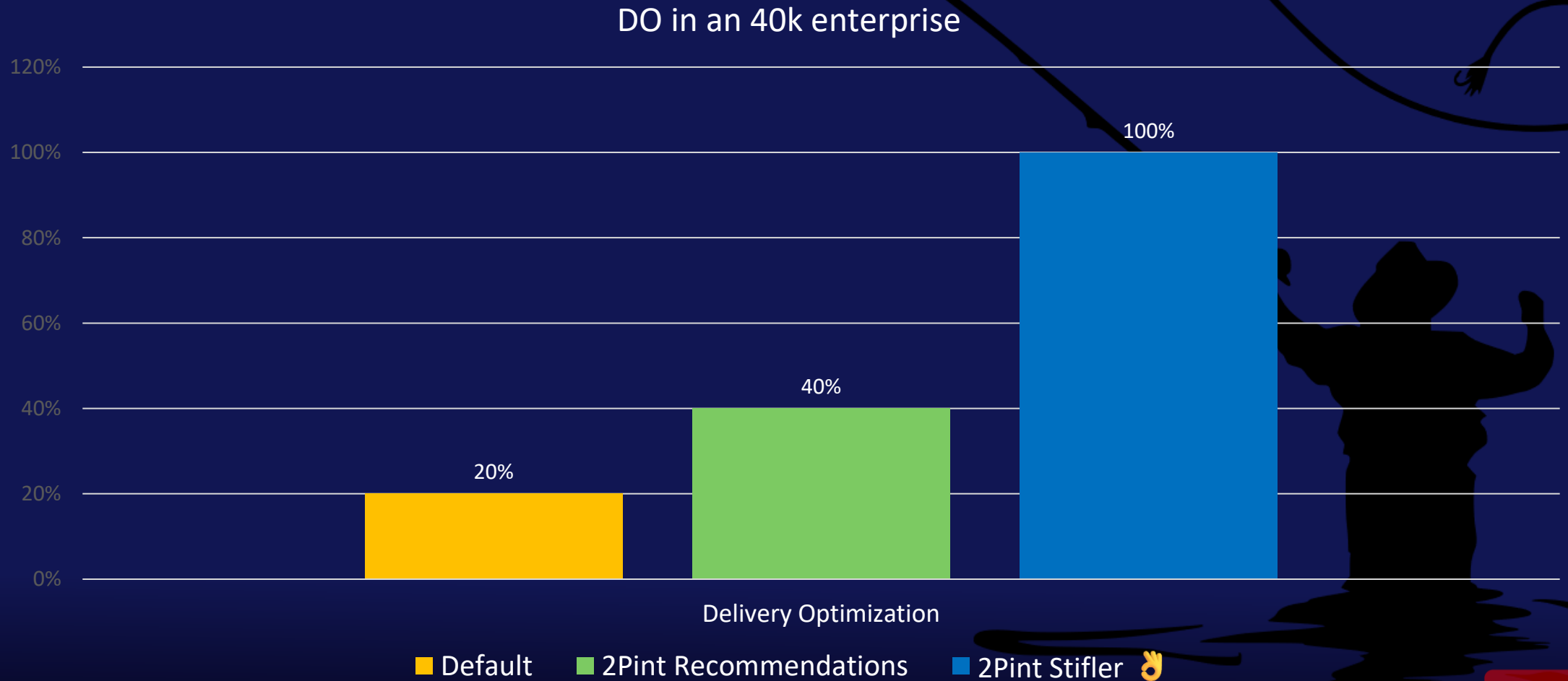
DEMO

Delivery Optimization

How to kick the tires...



Configuration Matters! There is not one for all!



*Excluding the first download from source

There is no simple “do this one thing”

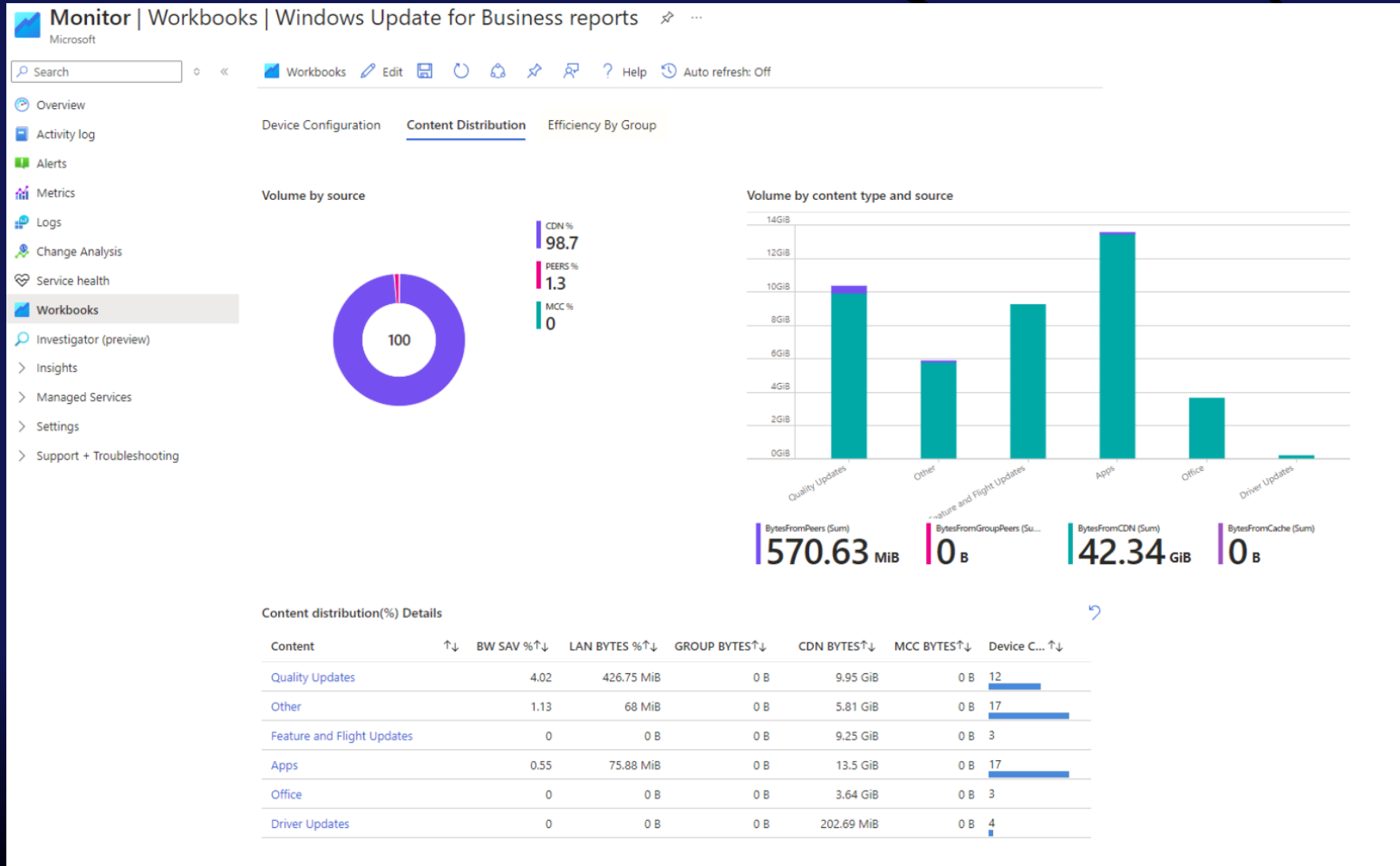
- ◆ Often need for different settings for different networks
- ◆ Some places are not built for peering
- ◆ Some places are not for caching
- ◆ Some places are not for downloads at all



Keeping an eye on things

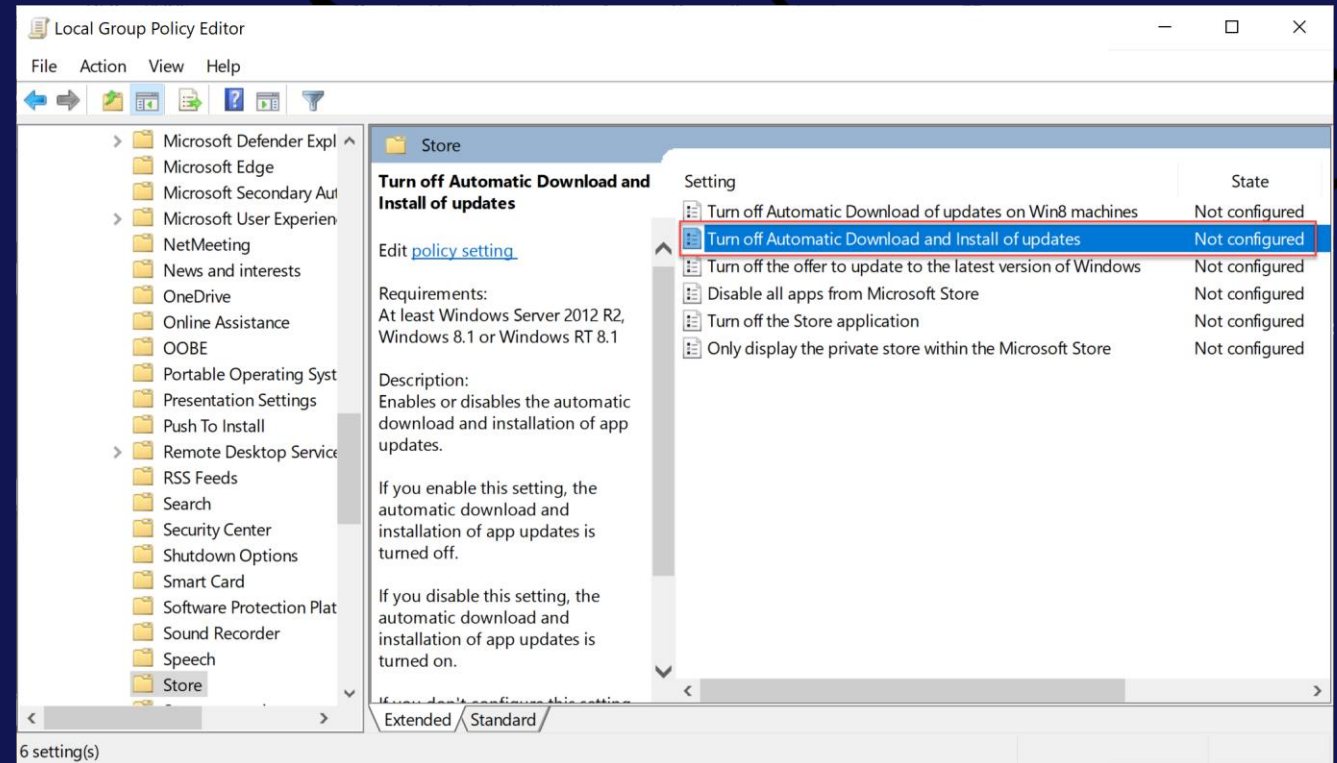


DO Report in Intune



General DO TIPS

- ◆ Know your Network! Buy the Network team a pint
- ◆ Make sure you have optimal P2P config for **your** network
 - ◆ [Delivery Optimization Recommendations for Microsoft Intune](#)
- ◆ Manage your Store Apps
 - ◆ Remove unwanted apps at build time
- ◆ Check endpoint connectivity
 - ◆ <https://t.ly/y5rd>
- ◆ Use the MS Troubleshooting script
 - ◆ <https://t.ly/e8AmP>



General DO TIPS 2

- ◆ If you have too many chunks of content in the cache it may result in bad peering
 - ◆ Choose 'Minimum file size to cache' and cache retention age carefully
<https://t.ly/294WT>
- ◆ For laptops, check the 'Allow uploads while device is on battery...etc' policy: If NOT enabled laptops will not use P2P while on battery as the default is 100%
- ◆ Use Intune policy to configure settings
- ◆ Do some basic testing and benchmarking for YOUR environment

Device Health Check:

-> Device Settings

Name	Result	Details
Download Mode	PASS	Internet
Service Status	PASS	dosvc running
Cache Folder Access	PASS	
Registry Key Access	PASS	
VPN	PASS	Not connected: 2Pint-VPN

-> Hardware Settings

Name	Result	Details
RAM	PASS	64 GB
Disk	PASS	C: Total Size: 1862.15GB Free Space: 1656.98GB
Power	PASS	A/C: Balanced

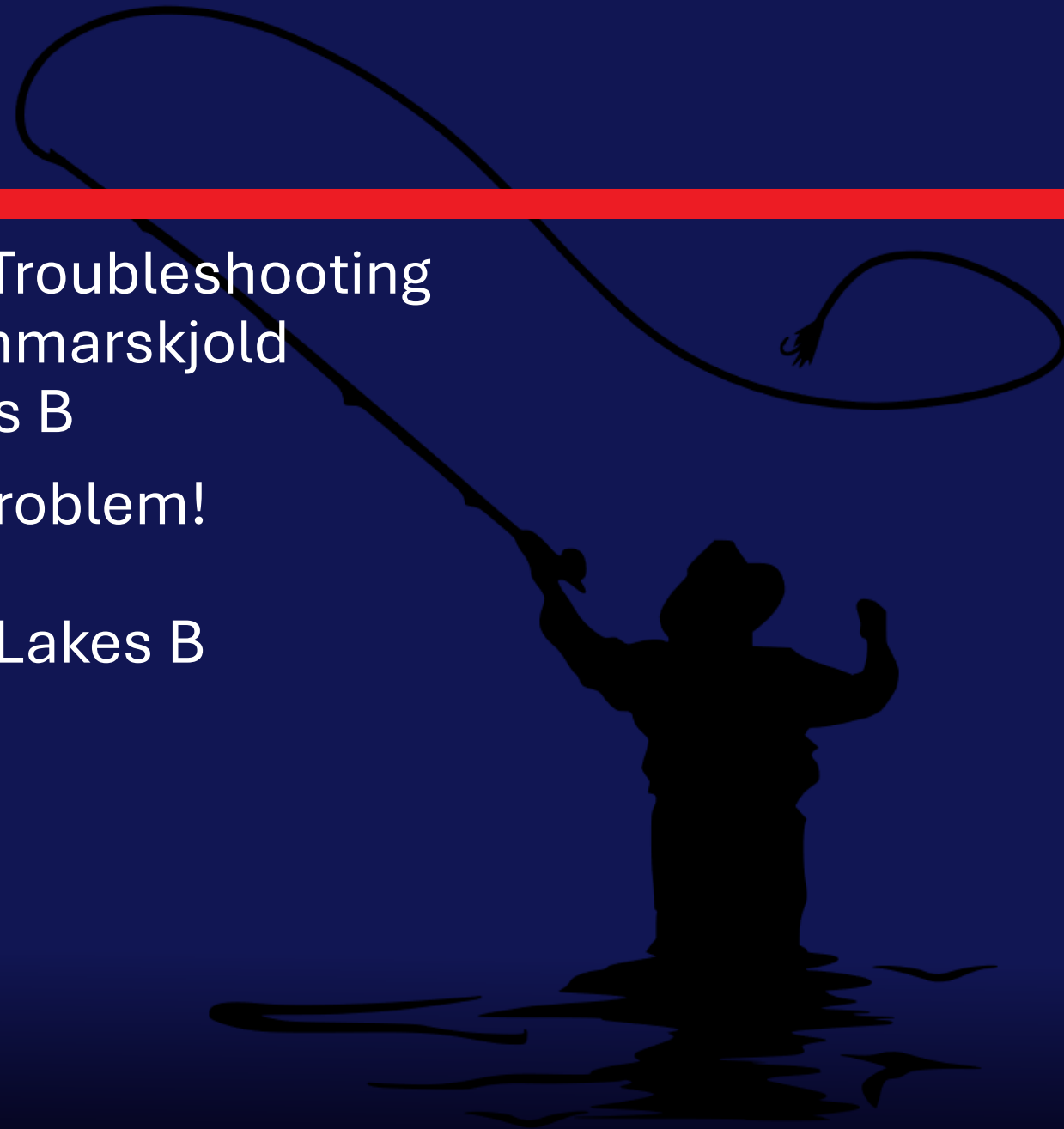
-> Connection Check

Name	Result	Details
Network Interface	PASS	(Ethernet) Realtek Gaming GbE Family Controller - (Wi-Fi) Intel(R) Wi-Fi 6 AX201 160MHz - (Bluetooth Network Connection) Bluetooth Device (Personal Area Network)
Check Port	PASS	7680
Check Port	WARN	3544
Internet Access	ERROR	Unable to reach DO's GEO service. Exception: System.Net.WebException: Unable to connect to the remote server ---> System.Net.Sockets.SocketException: An attempt was made to access a socket in a way forbidden by its access permissions 40.70.77.148:443 at System.Net.Sockets.Socket.InternalEndConnect(IAsyncResult asyncResult) at System.Net.Sockets.Socket.EndConnect(IAsyncResult asyncResult) at System.Net.ServicePoint.ConnectSocketInternal(Boolean connectFailure, Socket s4, Socket s6, Socket& socket, IPAddress& address, ConnectSocketState state, IAsyncResult asyncResult, Exception& exception) --- End of inner exception stack trace --- at Microsoft.PowerShell.Commands.WebRequestPSCmdlet.GetResponse(WebRequest request) at Microsoft.PowerShell.Commands.WebRequestPSCmdlet.ProcessRecord()
External IP	FAIL	Unable to get External IP.
HTTP Byte-Range Support	FAIL	Internet check failed. Unable to check HTTP Byte-Range Support
Host Connection	FAIL	Internet check failed. Unable to check dl.delivery.mp.microsoft.com
Host Connection	FAIL	Internet check failed. Unable to check download.windowsupdate.com

Peer_Info	Description
Peer to Peer Efficiency:	10.08187 %

Other Sessions

- ♦ Mastering Delivery Optimization Troubleshooting
Mattias Benninge & Andreas Hammarskjold
Tuesday, May 5th at 1:00 PM Lakes B
- ♦ Build your test lab: No Lab? No Problem!
Mattias Benninge & Mike Terill
Wednesday, May 6th at 17:00 PM Lakes B



For those who fell asleep... TL;DR;

- ◆ Look into Microsoft Connected Cache
- ◆ Https is incoming...
- ◆ Review your peering strategy
- ◆ Review the peering efficiency



Links worth saving

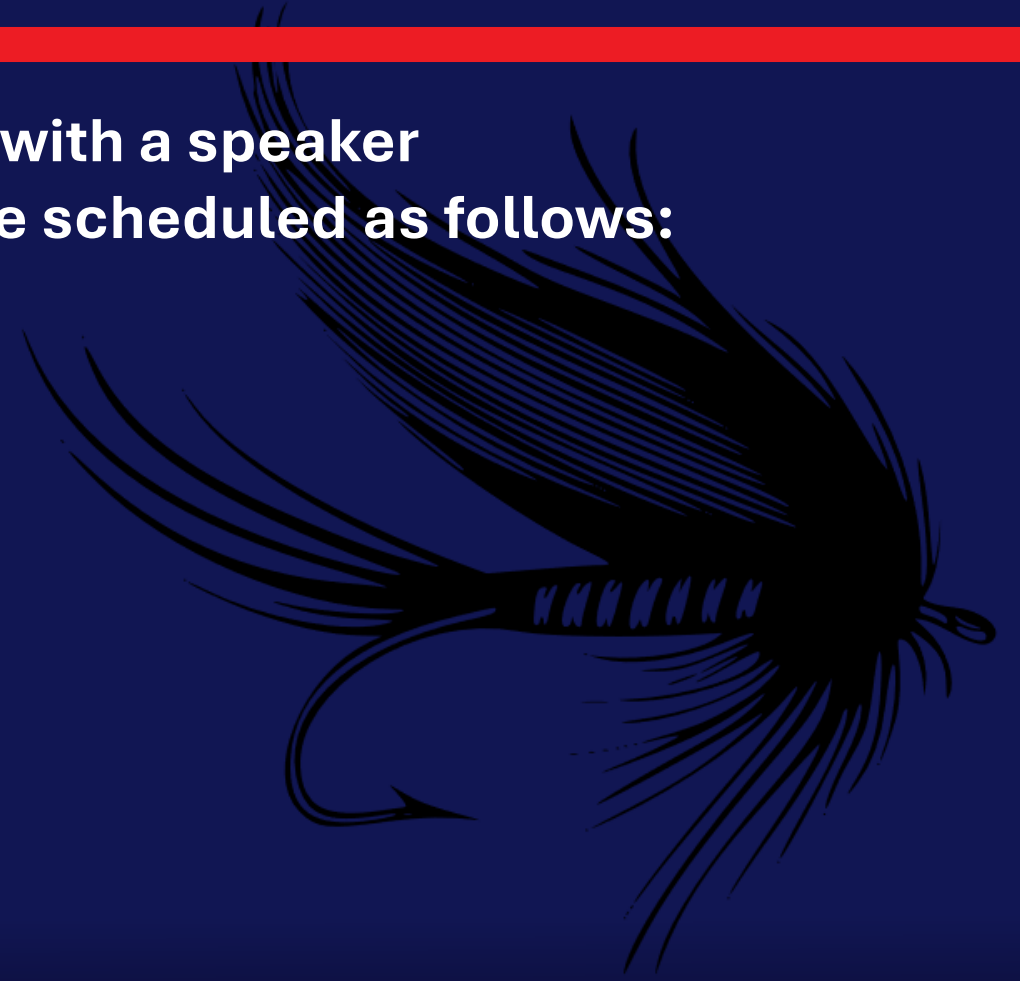
- ◆ <https://2pintsoftware.com/news/details/delivery-optimization-and-non-peerable-content>



Fishing Sessions

**Be sure to find Wally and sign up for 1:1 time with a speaker
If you'd like to "catch" us for a session, we are scheduled as follows:**

- ♦ Tuesday 15.00 – 16.40



We want your feedback!

MCC at MMS 2026 — Your
Feedback



SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



ninjaOne®



numecent®

